

27 | ENERGY, CARBON, AND SUSTAINABILITY TEAMS

80 practical questions for evidence-aware, privacy-preserving and technically defensible building decisions

- 1 Electricity, gas, heat and operational carbon
- 2 HVAC, environmental conditions and efficiency
- 3 Space use, occupancy context and service intensity
- 4 Travel, access and lower-impact choices
- 5 Water demand, leaks and conservation
- 6 Waste, materials, reuse and circularity
- 7 Procurement, projects and embodied carbon
- 8 Targets, baselines, reporting, scenarios and engagement

DECISION DISCIPLINE

1

VERIFY

Modality, location, mapping, commissioning, calibration or functional checks, clocks and current health

2

TRACE

Owner, precedence, effective/observed/retrieved times, permission, provenance, staleness and conflicts

3

ANALYSE

Operation type, decision interval, minimum evidence, method, uncertainty and safe failure

4

BOUND

No individual tracking, unsupported causality, certification, authority substitution or direct control

5

FAIL SAFELY

Decision-ready, caveated, directional, unknown or escalation - never fabricated certainty

HISTORICAL PLAN RULE

The May 2021 floor plan is a historical spatial reference only. It cannot establish current room use, access, services, hazards or closures.

Purpose, scope and catalogue map

A decision catalogue for energy, carbon and sustainability teams - not a claim that any capability is already deployed.

WHAT THIS CATALOGUE PROVIDES

Eighty natural stakeholder questions. Each links a real operational, reporting, investment, procurement or engagement decision to proposed privacy-preserving evidence, authoritative supporting data, an explicit analysis operation, a minimum evidence threshold, a strict answer boundary and a concrete action.

The service may provide a decision-ready answer, a caveated result, a directional signal, or an explicit unknown with the responsible owner and smallest safe evidence upgrade.

STATUS OF SENSORS, INTEGRATIONS AND ANALYTICS

Every sensor, integration and analytic method remains proposed until its deployment or collection method, physical location and current asset/zone mapping, commissioning, permission, calibration or functional verification, clock alignment and current health are independently checked.

A listed source is not proof that it exists, is accessible, is correctly mapped or is fit for the decision. Current authoritative owners and competent professionals always take precedence.

- NON-NEGOTIABLE BOUNDARIES**
 - No identification or tracking of individuals; no inference of behaviour, attendance, productivity, intent, responsibility or blame.
 - No causal claim from weak association and no silent substitution for authoritative meters, bills, finance records or approved reporting methods.
- No certification of carbon accounts, baselines, compliance, safety, accessibility, performance, savings or supplier claims.
 - No authorisation of access, budgets, projects, maintenance, procurement or reporting changes; no direct control of plant, equipment or building systems.

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How to read each question card

Six fields convert a stakeholder need into a bounded, traceable and decision-ready answer.

1

STAKEHOLDER QUESTION
A natural role-specific decision question, framed so that a useful answer can be acted on by a named owner.

2

SENSORS AND TELEMETRY
Proposed modality and physical location, spatial and temporal coverage, sampling and aggregation, commissioning, calibration or functional checks, clock alignment, uptime, gaps, drift and raw-to-aggregate provenance.

3

AUTHORITATIVE SUPPORTING DATA
Current source owners, precedence, effective/observed/retrieved times, permissions, version and provenance, staleness, conflicts and the resolver.

4

ANALYSIS REQUIRED
Operation type, decision interval, minimum evidence, method, uncertainty, alternative explanations and a safe-failure answer.

5

ANSWER BOUNDARY
What the service must not infer, disclose, promise, certify, authorise, change or control; which accountable owner retains authority.

6

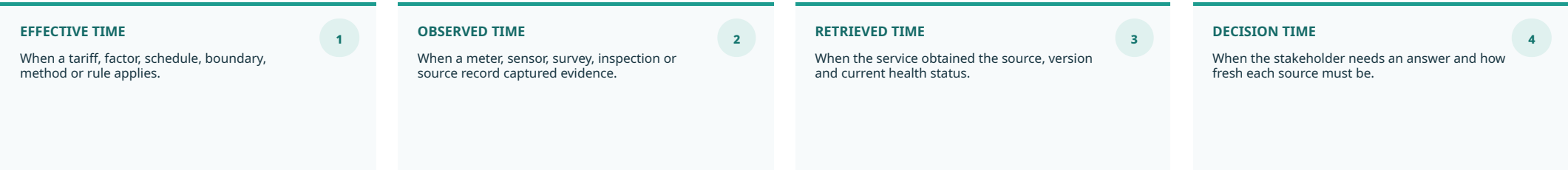
WHY THIS IS USEFUL
The concrete decision, prioritisation, escalation, trial, investment or evidence action enabled for the sustainability team.

ANALYSIS VOCABULARY

LOOKUP Retrieve an approved value or rule without transforming it.	OBSERVATION Describe measured or recorded evidence.	CALCULATION Apply deterministic arithmetic, allocation or reconciliation.	ESTIMATE Bound a quantity that is incomplete or indirectly observed.
FORECAST Project a future range under explicit assumptions.	COMPARISON Contrast equivalent periods, assets, zones or options.	DIAGNOSIS Rank candidate explanations; never prove cause from association alone.	RECOMMENDATION Propose an owner-controlled action only after evidence and authority gates pass.
PROPOSED Listed evidence or capability is not yet verified.	VERIFIED Method, mapping, permission and current health have passed the relevant checks.	DECISION-READY Minimum evidence, method, uncertainty and authority gates pass for the stated action.	

Evidence must be time-aware, traceable and safe to refuse

A plausible reading is not automatically a decision-ready answer.



EVIDENCE READINESS LADDER



PERMITTED ANSWER MODES

DECISION-READY	Evidence, method, permission and authority gates pass.
CAVEATED	Usable, but a bounded limitation materially affects interpretation.
DIRECTIONAL	Supports checking, measurement or option development - not certification or causal claims.
UNKNOWN / ESCALATE	A required source, owner, permission, quality, method or safety condition fails.

MINIMUM RELEASE CHECKS

- Sensor modality, location and asset/zone mapping suit the claim.
- Spatial and temporal coverage, aggregation and decision interval align.
- Units, direction, calibration or functional checks and clocks are known.
- Uptime, latency, gaps, substitutions, drift and provenance are visible.
- Authoritative owner, precedence, times, permissions and conflicts are explicit.
- Minimum evidence, uncertainty, alternatives and safe failure are shown.

1. Electricity, gas, heat and operational carbon

Category questions 1-2 of 10

SU-001CATEGORY 1ELECTRICITY, GAS, HEAT AND OPERATIONAL CARBON

STAKEHOLDER QUESTION

What reconciled electricity, gas and delivered-heat position is fit for the latest operational period, and what provisional operational carbon follows under the approved boundary and factor set?

SENSORS AND TELEMETRY

Evidence proposed. Fiscal electricity, gas and delivered-heat meters at verified site incomers, plus commissioned submeters for major plant and distribution branches. **Coverage.** Capture native intervals, normally 1-15 minutes, with units, direction, multiplier, clock and meter-health flags. **Aggregation.** Aggregate energy separately by fuel to half-hour, day and reporting period before any carbon conversion. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Estimated bills, unmapped circuits, meter exchange, resets, gaps and mixed meter classes can prevent a closed balance.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The estates energy owner controls the meter hierarchy, finance owns paid invoices and the authorised reporting owner controls boundaries and carbon-factor versions. For formal totals, finance-aligned fiscal data and approved reporting rules take precedence; BMS and temporary meters are diagnostic only. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale mapping, estimated reads and unresolved invoice-meter-factor differences; never select a convenient source silently. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Calculation and Comparison. **Decision interval.** Reconcile energy by fuel daily for operations and close each period only through the approved reporting process. **Minimum evidence.** Require verified meter identities, units, direction and boundary plus the locally approved completeness rule; below that threshold provide energy with a gap statement only. **Method and uncertainty.** Use balance reconciliation and versioned factors, keeping formal inventory carbon separate from any time-resolved operational indicator; quantify meter, estimation and factor uncertainty. **Safe failure.** Return energy-only, directional or unknown when the balance, factor purpose or reporting boundary is unresolved.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose restricted account or tariff detail, present a time-varying grid signal as a certified inventory figure, promise savings, certify accounts or compliance, authorise corrections or expenditure, or control meters, switchgear or plant. **Authority.** Finance-approved totals, the authorised method and responsible professionals retain authority.

WHY THIS IS USEFUL

Supports the operational period close, separates formal accounting from live indicators, and generates named meter, mapping or factor corrections.

SU-002CATEGORY 1ELECTRICITY, GAS, HEAT AND OPERATIONAL CARBON

STAKEHOLDER QUESTION

Which demand persists during verified closure or low-use windows, how much is explained by essential services, and which residual circuits need investigation?

SENSORS AND TELEMETRY

Evidence proposed. Main and end-use submeters covering overnight, weekends and declared closure windows. **Coverage.** Sample every 1-5 minutes across at least four comparable low-use periods. **Aggregation.** Aggregate stable 15-minute minima by fuel, circuit and closure type. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Essential, research, security or tenant loads may be omitted, while calendars and occupancy context may be incomplete.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The energy manager owns meter data and authorised service owners own the list of essential loads. Declared closures, approved exceptions and plant schedules take precedence over generic opening hours. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; conflicting closure or exception records must remain visible. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Observation, Calculation, Comparison, Diagnosis and Recommendation. **Decision interval.** Calculate minimum-load profiles and compare like-for-like low-use windows weekly and monthly. **Minimum evidence.** Require healthy meters, four valid windows and a current essential-load register. **Method and uncertainty.** Use robust medians, seasonal comparison and bounded subtraction of known essential loads; express a range for unexplained load rather than a single waste figure. **Safe failure.** Label the result unexplained when service context is missing.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose access logs, individual presence or sensitive essential-load detail, promise savings or target delivery, certify accounts, compliance or performance, authorise shutdowns, schedule changes or maintenance, or control circuits or plant. **Authority.** No safety-critical, research, security or environmental load may be treated as removable without its responsible owner.

WHY THIS IS USEFUL

Creates a circuit-level investigation list and strengthens closure and shutdown checklists before any load is challenged.

1. Electricity, gas, heat and operational carbon

Category questions 3-4 of 10

SU-003CATEGORY 1ELECTRICITY, GAS, HEAT AND OPERATIONAL CARBON

STAKEHOLDER QUESTION

Which half-hourly electrical peaks create capacity or tariff exposure, which major loads coincide, and which peaks are technically shiftable?

SENSORS AND TELEMETRY

Evidence proposed. Half-hourly main demand, 1-5 minute submeters for major plant and aggregate EV-charging load, plus non-identifying equipment-state points. **Coverage.** Synchronise clocks and capture tariff intervals continuously. **Aggregation.** Aggregate ranked peaks with coincident-load shares by day and billing period. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Missing submeters, demand ratchets, coincident starts and clock misalignment limit attribution.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The energy manager owns demand data and the finance tariff owner owns charging rules. Supplier settlement data take precedence over dashboard estimates; BMS and charger platforms provide context only. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; tariff, meter or timing conflicts require owner resolution. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Calculation, Estimate, Comparison and Recommendation. **Decision interval.** Rank peaks and decompose coincident major loads weekly and at each billing close. **Minimum evidence.** Require a current tariff and synchronised coverage of at least 80% of major loads. **Method and uncertainty.** Use contribution ranges and constrained counterfactual scheduling tests; express confidence by load and alternative explanations for residual demand. **Safe failure.** Report only verified peak time and magnitude when attribution is weak.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose vehicle, user or research identities and restricted tariff detail, promise savings or target delivery, certify accounts, compliance or performance, authorise load shedding, tariff commitments or schedule changes, or control chargers, circuits or plant. **Authority.** Safety, research and service constraints override optimisation suggestions.

WHY THIS IS USEFUL

Supports tariff assessment, capacity planning and owner-led start-time trials for only the peaks that are both material and flexible.

SU-004CATEGORY 1ELECTRICITY, GAS, HEAT AND OPERATIONAL CARBON

STAKEHOLDER QUESTION

Which utility or end-use profiles show a persistent step change, drift or repeating anomaly that merits a meter, controls or asset check?

SENSORS AND TELEMETRY

Evidence proposed. Continuous main and submeter streams with meter-health flags across utilities and major end uses. **Coverage.** Evaluate native intervals and retain maintenance, holiday and remapping markers. **Aggregation.** Aggregate hourly and daily profiles with change flags. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Resets, missing data, weather, operational changes and remapping can resemble genuine anomalies.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The meter asset owner controls commissioning and mapping status, while CMMS and BMS owners supply change context. Signed meter changes and verified work records take precedence over algorithmic flags. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; contradictory explanations must be shown, not suppressed. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Observation, Comparison, Diagnosis and Recommendation. **Decision interval.** Apply robust change-point, residual and peer-profile detection daily with a weekly triage cycle. **Minimum evidence.** Require 14-28 valid comparable days, a stable baseline and no unresolved meter intervention. **Method and uncertainty.** Use season-aware residuals and repeated-event scoring; express confidence, effect range and plausible non-fault causes. **Safe failure.** Issue an unverified anomaly for competent human checking.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose sensitive operational detail or any person's activity, promise savings or target delivery, certify accounts, compliance or performance, authorise fault declarations, work orders or data corrections, or control meters or BMS configuration. **Authority.** Pattern change alone is not evidence of theft, negligence or equipment failure.

WHY THIS IS USEFUL

Prioritises meter, controls or asset checks while distinguishing instrumentation problems from real consumption change.

1. Electricity, gas, heat and operational carbon

Category questions 5-6 of 10

SU-005CATEGORY 1ELECTRICITY, GAS, HEAT AND OPERATIONAL CARBON

STAKEHOLDER QUESTION

How much electricity can be allocated to major end uses, what remains unallocated, and which metering gap should be closed first?

SENSORS AND TELEMETRY

Evidence proposed. Commissioned circuit submeters, panel current transformers and asset power meters covering major end uses. **Coverage.** Sample at 1-15 minute intervals with polarity and health checks. **Aggregation.** Aggregate daily and monthly end-use totals plus an explicit unallocated residual. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Shared circuits, nonlinear loads, CT placement and stale panel schedules constrain disaggregation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Electrical services owns single-line diagrams and circuit schedules; the energy manager owns the meter hierarchy. Verified physical mapping takes precedence over labels or modelled allocations. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; drawing, label and measurement conflicts go to electrical services. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Observation, Calculation, Estimate, Comparison and Recommendation. **Decision interval.** Reconcile bottom-up end-use totals against the main meter monthly and after mapping changes. **Minimum evidence.** Require verified units and measured coverage of at least 70% of total consumption. **Method and uncertainty.** Use measured allocation first and bounded model estimates second; express allocation ranges and the unallocated share. **Safe failure.** Publish a large residual as unallocated rather than force an estimate.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose device-level activity, sensitive IT loads or research content, promise savings or target delivery, certify accounts, compliance or performance, authorise rewiring, submeter installation or reporting allocation, or control circuits, panels or devices. **Authority.** Estimated categories must never be presented as metered fact.

WHY THIS IS USEFUL

Shows where end-use action is possible now and which single submetering investment would most improve allocation confidence.

SU-006CATEGORY 1ELECTRICITY, GAS, HEAT AND OPERATIONAL CARBON

STAKEHOLDER QUESTION

After weather and service-hours adjustment, is gas or delivered heat changing materially, and which plant or control hypothesis should be tested first?

SENSORS AND TELEMETRY

Evidence proposed. Gas or heat meters, boiler or heat-interface telemetry, outdoor temperature and selected representative zone conditions. **Coverage.** Sample every 5-15 minutes through heating seasons. **Aggregation.** Aggregate daily fuel, degree-day, runtime and delivered-temperature indicators. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Meter lag, calorific conversion, thermal inertia, weather-source choice and partial zone coverage limit interpretation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The energy manager owns fuel totals; the BMS owner, weather-source owner and schedule owner supply context. Supplier invoices or heat-network billing take precedence over local calculations. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; factor, weather, schedule or meter conflicts require named-owner resolution. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Calculation, Comparison and Diagnosis. **Decision interval.** Fit weather- and schedule-aware energy models and test residual change weekly in season and monthly for reporting. **Minimum evidence.** Require one representative heating season or at least 30 valid comparable days. **Method and uncertainty.** Use change-point regression with service and maintenance markers; express prediction intervals and alternative explanations for residuals. **Safe failure.** State unresolved driver when evidence cannot separate causes.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose occupant activity, account detail or restricted plant conditions, promise savings or target delivery, certify accounts, compliance or performance, authorise set-point changes, plant work or reporting adjustments, or control boilers, heat interfaces or controls. **Authority.** Correlation with weather or hours does not establish a control defect.

WHY THIS IS USEFUL

Directs seasonal-tuning and plant investigation toward the strongest supported explanation rather than a raw year-on-year change.

1. Electricity, gas, heat and operational carbon

Category questions 7-8 of 10

SU-007CATEGORY 1ELECTRICITY, GAS, HEAT AND OPERATIONAL CARBON

STAKEHOLDER QUESTION

How much verified on-site generation was produced, used on site, exported, unavailable or curtailed, and which constraint best explains lost output?

SENSORS AND TELEMETRY

Evidence proposed. Generation inverter totals, import/export meters, storage state where present and approved curtailment flags. **Coverage.** Sample at 1-5 minutes and align with irradiance or weather context. **Aggregation.** Aggregate half-hour, daily and monthly generation, self-use and export balances. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Inverter clipping, outages, meter direction, storage losses and missing curtailment reasons constrain the balance.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The renewable-asset owner controls inverter and outage records; the energy manager controls import/export mapping. Settlement-grade import/export and certified generation meters take precedence over portal displays. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; unresolved direction, loss or outage conflicts remain explicit. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Observation, Calculation, Comparison and Diagnosis. **Decision interval.** Reconcile generation with import, export, storage and local demand daily with monthly closure. **Minimum evidence.** Require commissioned meter directions, aligned clocks and at least 90% valid intervals. **Method and uncertainty.** Use energy-balance checks plus weather-normalised expected-generation bands; express loss, curtailment and missing-data ranges. **Safe failure.** Report unallocated renewable energy when the balance does not close.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose tenant consumption, user behaviour or commercially sensitive account data, promise savings or target delivery, certify accounts, compliance or performance, authorise storage dispatch, export contracts or plant changes, or control inverters, batteries or switchgear. **Authority.** Do not state that low generation is a fault without competent inspection.

WHY THIS IS USEFUL

Supports renewable maintenance, self-consumption planning and transparent reporting of production, export and unavailable output.

SU-008CATEGORY 1ELECTRICITY, GAS, HEAT AND OPERATIONAL CARBON

STAKEHOLDER QUESTION

Which candidate energy investigations meet the materiality, persistence, ownership and evidence-readiness gates for the action pipeline?

SENSORS AND TELEMETRY

Evidence proposed. Validated meter trends, BMS runtime, environmental points and asset-health markers for candidate systems. **Coverage.** Use at least four weeks of representative operation and refresh candidate status weekly. **Aggregation.** Aggregate opportunity ranges by asset, service zone and intervention type. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Unmetered interactions, short baselines, seasonal change and maintenance activity may distort rankings.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The energy manager owns the opportunity register; asset and service owners control operational constraints. Measured, reconciled data and signed work records take precedence over dashboard estimates. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; conflicts about scope, condition or service need are routed to the responsible owner. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Observation, Calculation, Comparison, Diagnosis and Recommendation. **Decision interval.** Screen and rank opportunities by energy, carbon, persistence and evidence readiness weekly for triage and at each planning cycle. **Minimum evidence.** Require a verified baseline, mapped asset, named owner and at least three repeated occurrences. **Method and uncertainty.** Use multi-criteria scoring with sensitivity to uncertain savings and delivery constraints; express ranges and a separate evidence-readiness score. **Safe failure.** Place weak candidates in an evidence-gap queue, not the investment list.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose staff performance, occupant intent or restricted service detail, promise savings or target delivery, certify accounts, compliance or performance, authorise budgets, work programmes or operational changes, or control plant or controls. **Authority.** A ranked opportunity is not a guaranteed saving or a business case.

WHY THIS IS USEFUL

Moves only evidence-ready opportunities into engineering or business-case development and parks weak candidates in a measurement queue.

1. Electricity, gas, heat and operational carbon

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SU-009CATEGORY 1ELECTRICITY, GAS, HEAT AND OPERATIONAL CARBON

STAKEHOLDER QUESTION

Which owner-approved flexible loads could move toward lower operational grid-carbon periods without creating service, peak-demand or rebound penalties?

SENSORS AND TELEMETRY

Evidence proposed. An owner-approved operational grid-intensity feed joined to aggregate load and state points for assets whose flexibility has been documented. **Coverage.** Align each source to its supported interval, geography, publication latency and forecast horizon, and retain source-health flags. **Aggregation.** Aggregate feasible shift windows, avoided or added peak demand and modelled operational-carbon signals by asset class and day. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Forecast error, regional mismatch, hidden process constraints, rebound and an invalid load baseline can reverse the apparent benefit.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The reporting owner controls inventory factors, while the energy-operation owner controls any separate time-varying signal and each asset owner defines flexibility. Approved inventory factors, signed service constraints and safety limits take precedence over an optimisation result. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale forecasts, source-purpose mismatch and disputed flexibility; do not merge operational signals into formal carbon totals. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Estimate, Forecast, Comparison and Recommendation. **Decision interval.** Run constrained load-shift scenarios at the operational decision interval and reassess them when forecasts or service plans change. **Minimum evidence.** Require verified load mapping, a stable baseline, current flexibility limits and a valid operational factor for the stated geography and interval. **Method and uncertainty.** Use constrained scheduling with peak, rebound and service checks; report a range against a no-shift counterfactual and keep inventory accounting separate. **Safe failure.** Return no safe shift when any constraint, factor purpose or baseline is uncertain.

ANSWER BOUNDARY

Service boundary. Do not identify people, expose schedules or sensitive load identities, promise carbon savings, present modelled movement as delivered reduction, certify accounts, authorise flexibility contracts or operating changes, or control batteries, chargers or plant. **Authority.** Asset owners, service requirements and the authorised reporting method retain authority.

WHY THIS IS USEFUL

Identifies service-safe timing trials while preventing an operational grid signal from being mistaken for formal inventory carbon.

SU-010CATEGORY 1ELECTRICITY, GAS, HEAT AND OPERATIONAL CARBON

STAKEHOLDER QUESTION

Which meter-identity, period, unit, multiplier or invoice discrepancies must be resolved before the energy period can be closed?

SENSORS AND TELEMETRY

Evidence proposed. Fiscal meter reads, submeter totals, dashboard extracts and data-quality flags across the same meter identities. **Coverage.** Reconcile each billing period and after meter or supplier changes. **Aggregation.** Aggregate variance tables by meter, fuel, period and source. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Estimated bills, rollover, multipliers, unit conversion and boundary mismatch can create false differences.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Finance owns paid invoices, the energy manager owns meter identities and the reporting owner owns carbon treatment. Validated fiscal records take precedence for formal totals; local systems support diagnosis. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; unresolved source differences require joint owner sign-off before use. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Calculation, Comparison and Diagnosis. **Decision interval.** Perform serial, unit, multiplier, date-window and balance reconciliation monthly before formal reporting. **Minimum evidence.** Require matched meter identities, complete billing windows and documented estimates. **Method and uncertainty.** Use line-item reconciliation with tolerance bands and residual tracing; express confidence by source and an explicit unresolved variance. **Safe failure.** Withhold decision use when identity or period cannot be matched.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose account credentials, confidential tariffs or sensitive plant data, promise savings or target delivery, certify accounts, compliance or performance, authorise invoice correction, restatement or meter work, or control financial systems, meters or dashboards. **Authority.** The service must not overwrite finance records or silently select the convenient total.

WHY THIS IS USEFUL

Prevents period close, cost analysis and carbon decisions from using mismatched meters, units, dates or invoice assumptions.

2. HVAC, environmental conditions and efficiency

Category questions 1-2 of 10

SU-011CATEGORY 2HVAC, ENVIRONMENTAL CONDITIONS AND EFFICIENCY

STAKEHOLDER QUESTION

Which HVAC assets operate beyond approved service windows, and is each exception justified by safety, frost, specialist service, override or fault evidence?

SENSORS AND TELEMETRY

Evidence proposed. BMS commands, proof-of-run, valve or damper position and plant energy for air-handling, heating and cooling assets. Coverage. Sample every 1-5 minutes and compare with approved schedules and override events. Aggregation. Aggregate after-hours runtime by asset, zone served and reason code. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Missing proof points, local controllers, legitimate frost protection and undocumented exceptions limit classification.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The BMS owner controls point mapping and the service owner controls approved schedules and exceptions. Safety sequences and signed operational exceptions take precedence over generic schedules. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; mapping, schedule or exception conflicts go to those owners. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Comparison, Diagnosis and Recommendation. Decision interval. Classify out-of-window runtime and quantify repeated duration and energy daily with weekly action triage. Minimum evidence. Require verified command/proof pairs, current schedules and at least three repeated events. Method and uncertainty. Use event matching with exception filtering and weather context; express confidence in necessity and a bounded energy range. Safe failure. Label unexplained runtime and request owner confirmation.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose who was present, access history or sensitive service activity, promise savings or target delivery, certify accounts, compliance or performance, authorise schedule changes or work orders, or control HVAC plant or BMS points. Authority. Do not switch off frost, life-safety, laboratory or equipment-protection services.

WHY THIS IS USEFUL

Provides an exception list for plant and service owners to confirm, correct or retain before schedules are changed.

SU-012CATEGORY 2HVAC, ENVIRONMENTAL CONDITIONS AND EFFICIENCY

STAKEHOLDER QUESTION

Are HVAC start and stop times meeting required conditions by service start with minimal preheating, precooling and shutdown lag?

SENSORS AND TELEMETRY

Evidence proposed. Plant start/stop commands, representative zone temperature, outdoor conditions and authorised service windows. Coverage. Sample every 1-5 minutes through occupied and preconditioning periods. Aggregation. Aggregate warm-up, cool-down and post-service runtime by system and season. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Sparse zones, thermal lag, open windows, internal gains and changing service needs weaken inference.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The controls manager owns sequences and the service owner owns readiness requirements. Approved comfort, air-quality and safety requirements take precedence over energy timing. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; conflicting readiness, schedule or point mappings require specialist resolution. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Calculation, Estimate, Comparison and Recommendation. Decision interval. Estimate optimal-start residuals and post-service runtime weekly by season. Minimum evidence. Require 10 valid operating days, synchronised points and verified service zones. Method and uncertainty. Use weather-aware regression and percentile readiness checks; express prediction bands and sensitivity to zone selection. Safe failure. Issue a schedule assessment request rather than an automatic change.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose individual arrival patterns, room-user identities or restricted service requirements, promise savings or target delivery, certify accounts, compliance or performance, authorise start/stop edits or comfort standards, or control BMS schedules or plant. Authority. No recommendation may reduce verified ventilation, accessibility or safety provision.

WHY THIS IS USEFUL

Supports tighter optimum-start and stop settings while protecting service readiness and avoiding false comfort failures.

2. HVAC, environmental conditions and efficiency

Category questions 3-4 of 10

SU-013	CATEGORY 2	HVAC, ENVIRONMENTAL CONDITIONS AND EFFICIENCY
STAKEHOLDER QUESTION		
Where does heating and cooling occur at the same time in the same service zone, and is it a control conflict, legitimate mode or measurement artefact?		
SENSORS AND TELEMETRY	Evidence proposed. Heating and cooling valve commands, supply temperatures, plant energy and representative zone conditions for mapped systems. Coverage. Sample every 30-60 seconds where control hunting matters and otherwise every 5 minutes. Aggregation. Aggregate overlap events by zone, air-handling unit and control mode. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Reheat design, humidity control, frost protection, sensor bias and shared plant can make overlap legitimate.	
AUTHORITATIVE SUPPORTING DATA	Owners and precedence. The controls owner owns sequences and point mapping; mechanical services owns design intent. Verified design requirements and active safety modes take precedence over an efficiency flag. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; sequence, mapping or design conflicts require competent engineering interpretation. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.	
ANALYSIS REQUIRED	Operation type. Observation, Comparison and Diagnosis. Decision interval. Detect concurrent heating and cooling only where time, service zone, hydraulic or air-side boundary and control mode are verified, then assess repeated events daily and monthly. Minimum evidence. Require paired command and feedback points, current sequences, valid temperature evidence and at least three material events outside allowed modes. Method and uncertainty. Use state classification, temporal overlap and bounded energy estimation; show sensor bias, reheat, dehumidification, frost and specialist-service alternatives. Safe failure. Issue a pattern for competent controls checking, not a waste or defect declaration.	
ANSWER BOUNDARY	Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose sensitive specialist requirements, promise savings, certify performance, authorise sequence changes or remedial work, or control valves, dampers or plant. Authority. Legitimate reheat, humidity, frost, safety and specialist modes must remain visible and authoritative.	
WHY THIS IS USEFUL	Targets control-sequence investigation where simultaneous conditioning is persistent, material and not explained by a legitimate mode.	

SU-014	CATEGORY 2	HVAC, ENVIRONMENTAL CONDITIONS AND EFFICIENCY
STAKEHOLDER QUESTION		
Are heating and cooling supply-return temperatures, flow and valve states delivering useful delta-T at representative loads?		
SENSORS AND TELEMETRY	Evidence proposed. Calibrated supply and return temperatures, flow where available, pump state and plant energy at boilers, heat exchangers and chillers. Coverage. Sample every 30 seconds to 5 minutes during stable load periods. Aggregation. Aggregate delta-T distributions by plant, load band and season. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Sensor placement, calibration drift, low flow, bypasses and mixed circuits can distort values.	
AUTHORITATIVE SUPPORTING DATA	Owners and precedence. Mechanical services owns design ranges and the BMS owner controls point health. Commissioning records and calibrated instruments take precedence over dashboard thresholds. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; conflicting sensors or undocumented hydraulic changes require field confirmation. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.	
ANALYSIS REQUIRED	Operation type. Observation, Calculation, Comparison and Diagnosis. Decision interval. Compare paired supply-return temperatures only in stable, correctly directed flow and known operating modes, with weekly checks during active seasons. Minimum evidence. Require calibrated or field-validated sensors, confirmed flow direction, reliable pump or valve state and at least 20 representative events across useful load bands. Method and uncertainty. Use load-binned delta-T and heat-balance plausibility checks; show uncertainty from low flow, bypass, mixing, placement and sensor drift rather than applying one universal threshold. Safe failure. Request field measurement or hydraulic checking when the pair cannot support interpretation.	
ANSWER BOUNDARY	Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose restricted plant detail or inferred operator performance, promise savings or target delivery, certify accounts, compliance or performance, authorise set-point changes, balancing or component work, or control pumps, valves or plant. Authority. A low delta-T pattern alone does not identify its cause.	
WHY THIS IS USEFUL	Supports coil, flow, pump and control checks and helps distinguish low delta-T from low load or sensor error.	

2. HVAC, environmental conditions and efficiency

Category questions 5-6 of 10

SU-015CATEGORY 2HVAC, ENVIRONMENTAL CONDITIONS AND EFFICIENCY

STAKEHOLDER QUESTION

Is ventilation volume and fan energy proportionate to verified air-quality need, service context and specialist-space requirements?

SENSORS AND TELEMETRY

Evidence proposed. Fan power, speed, airflow or pressure, damper positions, CO2 and selected environmental points across mapped ventilation zones. **Coverage.** Sample every 1-5 minutes and preserve mode and alarm states. **Aggregation.** Aggregate fan energy and ventilation indicators by service window, zone and mode. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. CO2 is an imperfect occupancy proxy, airflow sensors may drift and specialist requirements may dominate.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Mechanical services owns ventilation standards and design intent; the BMS owner controls point mapping. Life-safety, health and specialist-space requirements take precedence over energy optimisation. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; airflow, sensor or requirement conflicts require competent verification. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Calculation, Comparison, Diagnosis and Recommendation. **Decision interval.** Relate fan energy and control response to verified ventilation demand indicators and approved service modes weekly, with seasonal assessment. **Minimum evidence.** Require commissioned airflow or pressure evidence, healthy environmental points and current design or specialist requirements; CO2 alone is never a sufficient denominator. **Method and uncertainty.** Use fan-law plausibility, mode comparison and bounded demand-control scenarios; show airflow, sensor and proxy uncertainty separately. **Safe failure.** State that proportionality is unknown when airflow or service evidence is weak.

ANSWER BOUNDARY

Service boundary. Do not identify people, infer occupancy from CO2 alone, disclose health or specialist-space information, promise savings, certify air quality or compliance, authorise ventilation reductions or control changes, or control fans, dampers or air-quality systems. **Authority.** Verified ventilation, infection-control, life-safety and specialist requirements retain precedence.

WHY THIS IS USEFUL

Guides ventilation tuning or verification without trading away indoor-air, specialist-space or safety requirements.

SU-016CATEGORY 2HVAC, ENVIRONMENTAL CONDITIONS AND EFFICIENCY

STAKEHOLDER QUESTION

Is heat recovery available and effective when conditions permit, and what bypass, fouling, frost or imbalance evidence explains lost recovery?

SENSORS AND TELEMETRY

Evidence proposed. Upstream and downstream air or water temperatures, flow or fan state, bypass position and associated power. **Coverage.** Sample every 1-5 minutes during valid recovery conditions. **Aggregation.** Aggregate effectiveness bands and bypass events by system and season. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Sensor placement, leakage, frost protection and unbalanced flows constrain effectiveness estimates.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Mechanical services owns design conditions and the controls owner owns bypass logic. Commissioning measurements and verified safety sequences take precedence over calculated effectiveness. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; sensor, flow or bypass conflicts require specialist checking. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Observation, Calculation, Comparison and Diagnosis. **Decision interval.** Calculate sensible or enthalpy effectiveness only when the installed arrangement, valid flow path, bypass state and stable boundary conditions are confirmed, then assess weekly in relevant seasons. **Minimum evidence.** Require healthy paired sensors, flow or fan-state evidence and at least 20 valid events spanning representative conditions. **Method and uncertainty.** Use bounded effectiveness formulas and comparison with commissioning expectations; show sensor, leakage and flow-imbalance uncertainty. **Safe failure.** Produce an inspection prompt without an efficiency value when validity conditions fail.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose restricted plant layouts or operator identities, promise savings or target delivery, certify accounts, compliance or performance, authorise cleaning, refurbishment or bypass changes, or control dampers, pumps or recovery equipment. **Authority.** Frost and safety sequences remain authoritative.

WHY THIS IS USEFUL

Prioritises heat-recovery cleaning, balancing, damper or frost-control checks and quantifies the plausible recoverable range.

2. HVAC, environmental conditions and efficiency

Category questions 7-8 of 10

SU-017CATEGORY 2HVAC, ENVIRONMENTAL CONDITIONS AND EFFICIENCY

STAKEHOLDER QUESTION

Which control loops show persistent hunting, saturation, override or actuator non-response with material energy or service impact?

SENSORS AND TELEMETRY

Evidence proposed. Command and feedback positions, set points, process values, override status and plant energy for mapped control loops. Coverage. Sample every 10-60 seconds where dynamics require it. Aggregation. Aggregate instability events by loop, duration, operating mode and energy context. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Network latency, quantisation, manual intervention and missing feedback can imitate instability.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The controls team owns loop configuration and override authority; asset owners supply operational context. Verified maintenance actions and safety overrides take precedence over algorithmic classification. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; command, feedback or override conflicts must be checked at source. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Observation, Comparison, Diagnosis and Recommendation. Decision interval. Detect oscillation, saturation and command-feedback divergence daily with a weekly controls queue. Minimum evidence. Require verified point pairs, stable clocking and at least three repeated events. Method and uncertainty. Use spectral or run-length indicators combined with mode-aware rules; express confidence and alternative causes such as sensor noise. Safe failure. Issue a suspected instability flag for specialist confirmation.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose operator identity, room-user activity or security-sensitive control details, promise savings or target delivery, certify accounts, compliance or performance, authorise loop tuning, override removal or work instructions, or control actuators, controllers or BMS commands. Authority. Responsible engineers decide whether observed behaviour is harmful or intentional.

WHY THIS IS USEFUL

Creates a ranked controls-maintenance list based on persistence, service effect and energy materiality.

SU-018CATEGORY 2HVAC, ENVIRONMENTAL CONDITIONS AND EFFICIENCY

STAKEHOLDER QUESTION

Which zones repeatedly consume HVAC energy without meeting approved environmental service expectations, and where does the likely fault boundary sit?

SENSORS AND TELEMETRY

Evidence proposed. Representative temperature, humidity, CO2, light-touch comfort feedback and zone-level HVAC energy across verified service zones. Coverage. Sample environmental points every 1-5 minutes and aggregate feedback to protected cohorts. Aggregation. Aggregate condition and energy distributions by zone, service period and season. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Sensor placement, local heat gains, open windows and sparse feedback reduce representativeness.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Facilities owns service standards; the energy manager owns energy allocation and the data-protection owner controls feedback use. Approved health, accessibility and specialist requirements take precedence over generic comfort bands. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; conflicting sensors, standards or zone mappings require owner action. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Observation, Calculation, Comparison, Diagnosis and Recommendation. Decision interval. Compare conditions and energy with service-specific bands and peer zones weekly and seasonally. Minimum evidence. Require commissioned sensors, current zone mapping and sufficient anonymous feedback or repeated measured exceedance. Method and uncertainty. Use distributional comparison rather than single averages; express coverage and representativeness uncertainty. Safe failure. Report an evidence gap rather than attribute discomfort or waste.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose health, disability, personal feedback or who occupied a space, promise savings or target delivery, certify accounts, compliance or performance, authorise set-point changes or service reductions, or control HVAC or environmental controls. Authority. No condition pattern may be used to judge occupants or staff.

WHY THIS IS USEFUL

Focuses diagnosis on zones with both high input and failed service outcome, reducing blind schedule changes.

2. HVAC, environmental conditions and efficiency

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SU-019	CATEGORY 2	HVAC, ENVIRONMENTAL CONDITIONS AND EFFICIENCY
STAKEHOLDER QUESTION		
Where do central schedules, terminal schedules, deadbands and local overrides conflict, and which responsible owner should reconcile them?		
SENSORS AND TELEMETRY	Evidence proposed. Read-only set points, schedules, mode values, command states and zone conditions across mapped control hierarchies. Coverage. Capture configuration snapshots after approved changes and sample operating values every 5 minutes. Aggregation. Aggregate conflict events by hierarchy, zone and service period. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Hidden local controllers, stale graphics and temporary overrides can obscure the active value.	
AUTHORITATIVE SUPPORTING DATA	Owners and precedence. The controls owner owns active configuration and signed change evidence. Current controller values and approved sequence documents take precedence over exported graphics or historical settings. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; unresolved configuration differences require source-level confirmation. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.	
ANALYSIS REQUIRED	Operation type. Lookup, Observation, Comparison, Diagnosis and Recommendation. Decision interval. Build a hierarchy-aware configuration comparison and test operational consequences after each approved change and monthly. Minimum evidence. Require verified controller hierarchy, synchronised points and current sequence documents. Method and uncertainty. Use rule consistency checks plus repeated-event evidence; express confidence in active-value attribution and effect range. Safe failure. Return a configuration conflict candidate, not an automatic correction.	
ANSWER BOUNDARY	Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose security-sensitive credentials, operator identities or restricted plant topology, promise savings or target delivery, certify accounts, compliance or performance, authorise configuration edits or sequence changes, or control controllers, thermostats or BMS settings. Authority. Change management and competent engineering approval remain mandatory.	
WHY THIS IS USEFUL	Gives controls and service owners a specific schedule, deadband or override conflict to reconcile.	

SU-020	CATEGORY 2	HVAC, ENVIRONMENTAL CONDITIONS AND EFFICIENCY
STAKEHOLDER QUESTION		
Which HVAC assets or service zones should be inspected, recommissioned or submetered first after combining energy impact, service risk and evidence readiness?		
SENSORS AND TELEMETRY	Evidence proposed. Combined energy, runtime, condition, alarm and representative environmental telemetry across major plant and zones. Coverage. Use a common seasonal window and refresh priority evidence monthly. Aggregation. Aggregate priority indicators by asset or zone with explicit coverage gaps. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Sparse sensing, shared service zones and maintenance activity limit fair comparison.	
AUTHORITATIVE SUPPORTING DATA	Owners and precedence. Mechanical services and the energy manager jointly own priorities; service owners define criticality. Safety, service criticality and verified maintenance evidence take precedence over energy ranking alone. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; priority conflicts require a joint owner decision. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.	
ANALYSIS REQUIRED	Operation type. Lookup, Observation, Calculation, Comparison, Diagnosis and Recommendation. Decision interval. Score energy materiality, service deviation, persistence, criticality and evidence readiness monthly and before programme planning. Minimum evidence. Require a mapped asset or zone, representative data and a named responsible owner. Method and uncertainty. Use multi-criteria ranking with sensitivity tests; express rank stability and missing-evidence penalties. Safe failure. Place low-confidence cases in an instrumentation queue.	
ANSWER BOUNDARY	Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose complainant identity, staff performance or sensitive service detail, promise savings or target delivery, certify accounts, compliance or performance, authorise budgets, work scope or operational changes, or control plant or controls. Authority. The ranking is a transparent starting point, not a work authorisation.	
WHY THIS IS USEFUL	Combines impact, risk and evidence readiness into a defensible HVAC inspection, recommissioning and metering programme.	

3. Space use, occupancy context and service intensity

Category questions 1-2 of 10

SU-021CATEGORY 3SPACE USE, OCCUPANCY CONTEXT AND SERVICE INTENSITY

STAKEHOLDER QUESTION

For comparable zones, what energy is used per verified serviced hour or privacy-safe use band, and is the denominator reliable enough for a decision?

SENSORS AND TELEMETRY

Evidence proposed. Mapped zone energy or a transparent allocation, authorised serviced hours and coarse anonymous occupancy bands only where privacy thresholds are met. **Coverage.** Capture occupancy proxies at their supported interval, suppress small counts and never retain device, credential or face identifiers. **Aggregation.** Aggregate monthly energy per serviced hour or broad occupancy band for comparable zone types, showing numerator and denominator coverage separately. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Shared systems, sensor undercount, booking no-shows, continuously served rooms and unstable allocation can make the ratio misleading.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The space manager owns current room use and booking rules; the energy manager owns zone allocation. Verified physical mapping and declared service hours take precedence over the May 2021 plan or inferred use. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; mapping, denominator or room-use conflicts require owner confirmation. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Calculation, Estimate and Comparison. **Decision interval.** Compute service-normalised energy monthly and by season only for comparable zones and only after testing numerator and denominator stability. **Minimum evidence.** Require verified zone-energy mapping, a lawful anonymous denominator, sufficient serviced hours and no small-count disclosure. **Method and uncertainty.** Use ratio analysis with shared-service allocation sensitivity and separate confidence grades for energy and use context. **Safe failure.** Publish energy and service context separately when normalisation is unstable or privacy thresholds fail.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose identities, attendance, exact movement or personal schedules, promise savings or target delivery, certify accounts, compliance or performance, authorise space allocation, metering or service changes, or control building services or access systems. **Authority.** No empty-looking period may be inferred from one proxy alone.

WHY THIS IS USEFUL

Provides a fair denominator for comparing serviced zones and blocks misleading intensity claims when use evidence is weak.

SU-022CATEGORY 3SPACE USE, OCCUPANCY CONTEXT AND SERVICE INTENSITY

STAKEHOLDER QUESTION

Which zones sustain high service energy during low declared use after shared systems, specialist loads and standby requirements are accounted for?

SENSORS AND TELEMETRY

Evidence proposed. Mapped zone energy, HVAC runtime, lighting state and privacy-preserving occupancy or booking bands. **Coverage.** Use half-hour aggregates over at least four representative weeks. **Aggregation.** Aggregate service demand and use bands by zone type and time block. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Shared plant, intermittent specialist activity, booked-but-unused periods and proxy error can misclassify zones.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Space management owns current use classification and authorised bookings; service owners define mandatory provision. Current signed use and service requirements take precedence over historical floor-plan labels. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; conflicting use, booking or energy allocations remain explicit. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Calculation, Comparison and Diagnosis. **Decision interval.** Run quadrant and peer-zone analysis for low use and high service demand monthly. **Minimum evidence.** Require verified zone mapping, representative weeks and current service categories. **Method and uncertainty.** Use robust percentile bands and exclusion of known exceptional periods; express uncertainty from proxy coverage and shared energy. **Safe failure.** Create a survey candidate, not a closure recommendation.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose who used or did not use a zone, attendance or research activity, promise savings or target delivery, certify accounts, compliance or performance, authorise closure, consolidation or schedule changes, or control space access, lighting or HVAC. **Authority.** Specialist, intermittent and inclusive-access uses must be considered by responsible owners.

WHY THIS IS USEFUL

Identifies zones for schedule, standby-load or shared-system investigation without treating low declared use as proof of waste.

3. Space use, occupancy context and service intensity

Category questions 3-4 of 10

SU-023	CATEGORY 3	SPACE USE, OCCUPANCY CONTEXT AND SERVICE INTENSITY
STAKEHOLDER QUESTION		
How do approved teaching, events, cleaning, maintenance and closure periods change aggregate energy demand and plant runtime?		
SENSORS AND TELEMETRY	Evidence proposed. Aggregate interval energy, plant state and non-identifying activity categories linked to approved schedules. Coverage. Use before, during and after windows with at least 15-minute granularity. Aggregation. Aggregate incremental demand ranges by declared service category and season. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Overlapping activities, shared systems and timing errors limit attribution.	
AUTHORITATIVE SUPPORTING DATA	Owners and precedence. Timetabling, events, cleaning and maintenance owners control declared periods; the energy manager owns meter mapping. Signed operational records take precedence over inferred activity. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; overlapping or contradictory schedules must be retained as ambiguity. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.	
ANALYSIS REQUIRED	Operation type. Lookup, Observation, Calculation, Comparison and Diagnosis. Decision interval. Perform matched-window comparison for each declared service category monthly and after major events. Minimum evidence. Require at least five comparable occurrences, valid baseline windows and mapped services. Method and uncertainty. Use weather- and day-type adjusted difference ranges; express overlap and baseline uncertainty. Safe failure. Report combined service effect when categories cannot be separated.	
ANSWER BOUNDARY	Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose participant identities, attendance, staff activity or event content, promise savings or target delivery, certify accounts, compliance or performance, authorise service restrictions, charging or schedule changes, or control building services. Authority. The result describes declared service categories, not people.	
WHY THIS IS USEFUL	Helps plan plant schedules and service coverage for recurring activity types rather than generic opening hours.	

SU-024	CATEGORY 3	SPACE USE, OCCUPANCY CONTEXT AND SERVICE INTENSITY
STAKEHOLDER QUESTION		
Could compatible activities be consolidated by zone or time to reduce active service areas while preserving access, capacity and specialist constraints?		
SENSORS AND TELEMETRY	Evidence proposed. Zone energy, environmental service runtime, anonymous capacity bands and authorised booking capacity for candidate spaces. Coverage. Use representative half-hour and seasonal aggregates. Aggregation. Aggregate scenario energy, carbon and capacity by proposed consolidation pattern. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Future demand, room suitability, shared services and accessibility constraints may be uncertain.	
AUTHORITATIVE SUPPORTING DATA	Owners and precedence. Space management owns suitability and bookings; accessibility and service owners control mandatory constraints. Current room status and competent accessibility decisions take precedence over historical plan assumptions. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; conflicting capacity, access or service evidence must stop scenario release. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.	
ANALYSIS REQUIRED	Operation type. Lookup, Estimate, Forecast, Comparison and Recommendation. Decision interval. Run constrained consolidation scenarios against verified demand and service limits at each timetabling or space-planning cycle. Minimum evidence. Require current suitability, accessible capacity, representative demand and mapped energy. Method and uncertainty. Use scenario comparison with no-overflow and service constraints; express range for attendance and shared-energy response. Safe failure. Return infeasible or evidence insufficient rather than recommend a move.	
ANSWER BOUNDARY	Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose who attends, individual timetables or disability information, promise savings or target delivery, certify accounts, compliance or performance, authorise room moves, closures or timetable decisions, or control access, booking or building services. Authority. Human decision-makers must assess pedagogy, welfare, inclusion and specialist requirements.	
WHY THIS IS USEFUL	Supports an owner-approved consolidation trial with explicit accessibility, capacity and specialist-service safeguards.	

3. Space use, occupancy context and service intensity

Category questions 5-6 of 10

SU-025

CATEGORY 3

SPACE USE, OCCUPANCY CONTEXT AND SERVICE INTENSITY

STAKEHOLDER QUESTION

Which operational baseline best represents current space and service conditions for near-term decisions, and how sensitive is it to use and context assumptions?

SENSORS AND TELEMETRY

Evidence proposed. Validated utility and submeter data, outdoor conditions, approved schedules and anonymous aggregate use indicators. **Coverage.** Use daily and monthly data over a representative pre-period. **Aggregation.** Aggregate baseline predictions and residuals by fuel and reporting period. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Structural changes, missing periods, severe weather and low-use disruption can invalidate a baseline.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The energy and reporting owners approve baseline rules; service and space owners identify material changes. Approved methods and reconciled actuals take precedence over a convenient historical average. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; method, boundary or documented-change differences are resolved by the reporting owner. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Calculation, Estimate and Comparison. **Decision interval.** Fit weather- and service-aware baseline models and test stability before target setting and at least annually. **Minimum evidence.** Require 12 representative months or a justified alternative with sufficient seasonal coverage. **Method and uncertainty.** Use change-point regression with holdout testing and documented exclusions; express prediction intervals and sensitivity to excluded periods. **Safe failure.** Declare no stable baseline when structural change dominates.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose individual presence, staff performance or confidential service activity, promise savings or target delivery, certify accounts, compliance or performance, authorise baseline approval, restatement or target setting, or control source records or services. **Authority.** Models must not override finance or reporting records.

WHY THIS IS USEFUL

Provides a current operational comparator for space-service decisions while keeping it separate from the approved reporting baseline.

SU-026

CATEGORY 3

SPACE USE, OCCUPANCY CONTEXT AND SERVICE INTENSITY

STAKEHOLDER QUESTION

Which specialist or research-support loads remain elevated outside authorised activity windows, and what owner context is needed before escalation?

SENSORS AND TELEMETRY

Evidence proposed. Approved dedicated asset or circuit power meters, equipment run-state and authorised availability windows. **Coverage.** Sample at 1-5 minutes and report only asset-class or protected-zone aggregates. **Aggregation.** Aggregate outside-window energy and persistence by owner-approved asset group. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Shared circuits, standby needs, confidentiality and undocumented support activity limit interpretation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Equipment owners control operational requirements and permission; electrical services controls circuit mapping. Manufacturer constraints, safety needs and signed availability requirements take precedence over generic shutdown assumptions. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; mapping or requirement conflicts go to the equipment owner. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Comparison and Diagnosis. **Decision interval.** Compare metered demand with owner-declared operating and standby windows weekly for persistent loads. **Minimum evidence.** Require verified asset mapping, current requirements and at least three repeated outside-window events. **Method and uncertainty.** Use event classification with baseline standby bands; express range for avoidable share and missing-context flags. **Safe failure.** Label unexplained specialist load and request owner context.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose research content, user identity, device communications or experimental schedules, promise savings or target delivery, certify accounts, compliance or performance, authorise shutdowns, policy changes or maintenance, or control specialist circuits or equipment. **Authority.** Safety, data integrity and equipment protection take precedence.

WHY THIS IS USEFUL

Routes unexplained specialist loads to the correct service owner and defines the context needed before an engineering check.

3. Space use, occupancy context and service intensity

Category questions 7-8 of 10

SU-027CATEGORY 3SPACE USE, OCCUPANCY CONTEXT AND SERVICE INTENSITY

STAKEHOLDER QUESTION

For approved after-hours use, what minimum HVAC, lighting, access and support package is required, and where are unnecessarily broad service zones activated?

SENSORS AND TELEMETRY

Evidence proposed. Aggregate entry counts or bookings, approved zone service requests and plant or lighting runtime. Coverage. Use 15-minute aggregates during authorised after-hours periods. Aggregation. Aggregate minimum served zones, active systems and associated energy by event type. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Tailgating, remote work, safety patrols and shared plant can obscure true demand.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Security and space owners confirm authorised windows; plant owners define minimum service and exceptions. Life-safety, security and equipment-protection requirements take precedence over energy reduction. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; authorisation, service or mapping conflicts require human confirmation. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Calculation, Estimate, Comparison and Recommendation. Decision interval. Map authorised demand to active service zones and quantify avoidable distribution weekly and before after-hours policy changes. Minimum evidence. Require current access rules, mapped systems and at least four comparable events. Method and uncertainty. Use service-zone intersection and constrained scenario analysis; express uncertainty in attendance and minimum-service needs. Safe failure. Return a zoning study request when demand cannot be safely isolated.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose access histories, names or expected users, promise savings or target delivery, certify accounts, compliance or performance, authorise after-hours access, service levels or plant schedules, or control lighting, HVAC or security systems. Authority. Emergency and welfare requirements cannot be reduced by the service.

WHY THIS IS USEFUL

Supports smaller after-hours service zones and more precise service packages without denying authorised access or support.

SU-028CATEGORY 3SPACE USE, OCCUPANCY CONTEXT AND SERVICE INTENSITY

STAKEHOLDER QUESTION

Is current anonymous space-use evidence strong enough to change an HVAC or lighting schedule, or is a bounded trial or extra measurement safer?

SENSORS AND TELEMETRY

Evidence proposed. Privacy-preserving counts, booking indicators, door-state aggregates and service requests for mapped zones. Coverage. Summarise completeness, latency and agreement weekly. Aggregation. Aggregate evidence-readiness scores by zone and schedule window. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Each proxy has blind spots, and absence of a signal is not proof of absence.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Space management owns authorised use; sensor and service owners define data fitness and permissions. Declared service requirements take precedence over proxy inference. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; disagreement between proxies or stale room mappings blocks schedule use. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Comparison, Diagnosis and Recommendation. Decision interval. Score coverage, agreement, temporal alignment and representativeness before each schedule decision. Minimum evidence. Require at least two independent permitted sources or one validated source with strong coverage. Method and uncertainty. Use evidence triangulation with false-negative checks; express confidence and explicit blind spots. Safe failure. Request measurement or owner confirmation instead of a schedule change.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose identity, attendance, exact location or personal routine, promise savings or target delivery, certify accounts, compliance or performance, authorise schedule changes or occupancy policy, or control lighting, HVAC or access systems. Authority. A missing signal must never be treated as proof that a space is empty.

WHY THIS IS USEFUL

Prevents schedule changes based on weak proxies and specifies the smallest trial or measurement needed for a safe decision.

3. Space use, occupancy context and service intensity

Category questions 9-10 of 10

SU-029

CATEGORY 3

SPACE USE, OCCUPANCY CONTEXT AND SERVICE INTENSITY

STAKEHOLDER QUESTION

Which comparable space groups are persistent outliers in energy per declared service unit, and which technical explanations should be checked first?

SENSORS AND TELEMETRY

Evidence proposed. Zone energy, authorised service hours, capacity, equipment inventory and anonymous use bands. **Coverage.** Aggregate monthly across comparable room or service types. **Aggregation.** Aggregate energy intensity by declared service unit with peer bands. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Allocation error; legacy equipment, fabric, exposure and service intensity can dominate the ratio.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Space and energy managers jointly own classification and allocation. Current service categories, verified equipment and mapped energy take precedence over generic area benchmarks. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; classification or allocation conflicts must be shown. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Calculation, Comparison and Diagnosis. **Decision interval.** Benchmark only comparable groups and decompose major structural differences monthly or termly. **Minimum evidence.** Require at least five peers or a longitudinal baseline and verified denominators. **Method and uncertainty.** Use robust peer distributions with weather and service covariates; express small-group and allocation uncertainty. **Safe failure.** State that comparison is not valid when peers are insufficient.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose occupant effort, teaching quality, productivity or blame, promise savings or target delivery, certify accounts, compliance or performance, authorise space ranking, service reduction or investment, or control building services. **Authority.** Results are for technical investigation, not judgement of people or teams.

WHY THIS IS USEFUL

Creates a peer-comparison list for controls, loads, schedules and mapping checks rather than a league table of occupants.

SU-030

CATEGORY 3

SPACE USE, OCCUPANCY CONTEXT AND SERVICE INTENSITY

STAKEHOLDER QUESTION

Before approval, how would a proposed timetable, room-use or layout change affect active service zones, peaks, energy and operational carbon across plausible scenarios?

SENSORS AND TELEMETRY

Evidence proposed. Historical zone energy, HVAC and lighting response joined to the approved proposal and current service-zone maps. **Coverage.** Use representative half-hour profiles and seasonal cases. **Aggregation.** Aggregate scenario ranges for energy, carbon, peak demand and active service zones. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Future behaviour, construction effects, shared systems and incomplete mapping make scenarios uncertain.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The change owner supplies the approved proposal; space, accessibility, fire and energy owners control constraints. Current competent decisions and verified service maps take precedence over the May 2021 plan. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; any unresolved access, capacity, safety or mapping conflict blocks recommendation. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Estimate, Forecast, Comparison and Recommendation. **Decision interval.** Run calibrated before-and-after service scenarios at each design or timetabling gate. **Minimum evidence.** Require a defined proposal, current constraints and representative baseline data. **Method and uncertainty.** Use profile substitution with shared-system and peak-demand checks; express scenario ranges and sensitivity to occupancy and controls response. **Safe failure.** Present assumptions only when calibration is weak.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose expected individual movement or personal timetable, promise savings or target delivery, certify accounts, compliance or performance, authorise the change, room allocation or capital commitment, or control booking or building services. **Authority.** Scenarios are decision support, not commitments or approvals.

WHY THIS IS USEFUL

Lets timetable, space and energy owners compare options before approval and see peak, service-zone and carbon trade-offs.

4. Travel, access and lower-impact choices

Category questions 1-2 of 10

SU-031CATEGORY 4TRAVEL, ACCESS AND LOWER-IMPACT CHOICES

STAKEHOLDER QUESTION

Which aggregate travel-mode evidence is representative enough for planning, and which periods or broad user groups remain under-represented?

SENSORS AND TELEMETRY

Evidence proposed. Voluntary anonymous travel surveys, aggregate pedestrian and cycle counters, and coarse occupancy of cycle and parking facilities without number plates, faces or device identifiers. **Coverage.** Collect in declared sampling windows across terms and seasons, documenting response and counter coverage. **Aggregation.** Aggregate mode-share ranges by broad user group, visit purpose and period only where group sizes pass privacy thresholds. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Self-selection, non-response, shared destinations, weather and event mix can make the sample unrepresentative.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The transport planning owner controls survey definitions and permissions; property and estates owners supply aggregate counts. Published institutional travel methods take precedence over informal mode estimates. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; conflicting samples or definitions must be reconciled by the transport owner. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Estimate and Comparison. **Decision interval.** Estimate aggregate mode share at each approved survey wave and test representativeness before comparing periods. **Minimum evidence.** Require a declared sampling frame, minimum group size, response-rate information and at least two complementary evidence sources for high-stakes decisions. **Method and uncertainty.** Use weighted survey estimates with count-based sense checks; show non-response, seasonal and classification uncertainty. **Safe failure.** Report only the observed sample and missing groups when population inference is not defensible.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose journeys, home locations, vehicle identities or personal circumstances, promise savings or target delivery, certify accounts, compliance or performance, authorise travel policy, incentives or infrastructure investment, or control access or transport systems. **Authority.** No aggregate mode result may be used to infer an individual's choices.

WHY THIS IS USEFUL

Provides a defensible travel-planning baseline and identifies where another survey wave or complementary count is needed.

SU-032CATEGORY 4TRAVEL, ACCESS AND LOWER-IMPACT CHOICES

STAKEHOLDER QUESTION

When do cycle parking, lockers and showers reach usable capacity, and is maintenance, rebalancing or expansion justified?

SENSORS AND TELEMETRY

Evidence proposed. Non-imaging rack occupancy, aggregate locker availability, shower water-use signals and anonymous fault reports. **Coverage.** Sample every 5-15 minutes and report occupancy bands. **Aggregation.** Aggregate daily peaks, dwell bands and seasonal capacity indicators. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Abandoned cycles, shared facilities, sensor obstruction and water use unrelated to cycling limit interpretation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Travel and estates teams own facility capacity and status; maintenance records define outages. Current accessibility, security and maintenance constraints take precedence over sensor availability. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; capacity, outage or sensor conflicts require site confirmation. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Observation, Calculation, Comparison, Diagnosis and Recommendation. **Decision interval.** Compare peak demand bands with usable capacity and outage-adjusted supply monthly and before investment decisions. **Minimum evidence.** Require healthy sensors, verified usable capacity and at least four representative weeks. **Method and uncertainty.** Use percentile demand and persistence analysis; express coverage and classification uncertainty. **Safe failure.** Request a manual count when sensors cannot distinguish usable capacity.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose cyclist identity, gender, work pattern or shower behaviour, promise savings or target delivery, certify accounts, compliance or performance, authorise facility expansion, access rules or maintenance priorities, or control doors, lockers or water systems. **Authority.** Accessibility, safeguarding and security requirements take precedence.

WHY THIS IS USEFUL

Directs maintenance and capacity investment to the facility and time window where usable supply is persistently constrained.

4. Travel, access and lower-impact choices

Category questions 3-4 of 10

SU-033CATEGORY 4TRAVEL, ACCESS AND LOWER-IMPACT CHOICES

STAKEHOLDER QUESTION

How do EV chargers contribute to feeder demand, coincidence, utilisation and failed access, and which capacity or policy issue needs owner assessment?

SENSORS AND TELEMETRY

Evidence proposed. Aggregate charger status, session energy and duration, feeder demand and availability counts without account or user identifiers. **Coverage.** Capture 1-15 minute intervals and retain failed-session counts. **Aggregation.** Aggregate utilisation, queue proxy, coincident peak and energy by charger group. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Plugged-but-not-charging states, platform outages and shared feeder limits constrain interpretation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The charger owner controls operational data; electrical services owns feeder capacity and finance owns tariff rules. Verified feeder studies and current tariff or access policy take precedence over dashboard estimates. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; session, feeder or tariff conflicts require owner resolution. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Calculation, Comparison, Diagnosis and Recommendation. **Decision interval.** Calculate utilisation, coincidence and constrained charging scenarios monthly and at capacity-planning gates. **Minimum evidence.** Require verified charger inventory, feeder mapping and at least four representative weeks. **Method and uncertainty.** Use interval coincidence and queue-sensitive scenario modelling; express range for future demand and managed-charging response. **Safe failure.** Return a capacity study need when electrical evidence is incomplete.

ANSWER BOUNDARY

Service boundary. Do not identify drivers or vehicles, expose account or journey detail, infer attendance or behaviour, promise emissions reduction, certify charging performance, authorise access or tariff changes, or control chargers or site load. **Authority.** The charger owner, electrical-capacity owner and approved access policy retain authority.

WHY THIS IS USEFUL

Supports charger maintenance, feeder-capacity assessment and policy discussion without exposing drivers, vehicles or journeys.

SU-034CATEGORY 4TRAVEL, ACCESS AND LOWER-IMPACT CHOICES

STAKEHOLDER QUESTION

Which building-related fleet or service duty classes are technically suitable for lower-carbon alternatives without compromising operational need?

SENSORS AND TELEMETRY

Evidence proposed. Approved fleet fuel totals, aggregate charger energy, odometer or service records and route-class summaries generated by the fleet owner. **Coverage.** Report monthly without continuous individual or public-route tracking. **Aggregation.** Aggregate energy and carbon by vehicle class, duty type and service period. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Shared vehicles, incomplete mileage, external charging and duty differences limit comparison.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The fleet manager owns duty definitions and vehicle records; finance owns fuel purchases and the reporting owner owns factors. Verified fleet and finance records take precedence over inferred mileage. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; vehicle, fuel or duty conflicts remain with the fleet owner. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Estimate, Comparison and Recommendation. **Decision interval.** Calculate energy and carbon intensity by authorised duty class and test feasible alternatives quarterly and at replacement planning. **Minimum evidence.** Require matched fuel or energy, mileage and duty data for a representative period. **Method and uncertainty.** Use range-based total-cost and carbon scenarios with service constraints; express duty-cycle, charging and utilisation uncertainty. **Safe failure.** State no comparable alternative when service requirements are unmatched.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose driver identity, exact routes, performance or location history, promise savings or target delivery, certify accounts, compliance or performance, authorise vehicle replacement, duty allocation or procurement, or control vehicles or chargers. **Authority.** Operational and safety requirements take precedence.

WHY THIS IS USEFUL

Targets replacement studies at duty classes with a credible technical alternative and avoids forcing unsuitable operational journeys.

4. Travel, access and lower-impact choices

Category questions 5-6 of 10

SU-035CATEGORY 4TRAVEL, ACCESS AND LOWER-IMPACT CHOICES

STAKEHOLDER QUESTION

Which published lower-impact routes and interchanges are currently usable, accessible and open, and where must guidance be caveated?

SENSORS AND TELEMETRY

Evidence proposed. Public transport status feeds, lift or accessible-door availability, route obstruction reports and weather context. Coverage. Refresh at source-supported intervals before common arrival periods. Aggregation. Aggregate current route-option status by broad access need, not person. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Coverage, latency, temporary works and unreported barriers can make status incomplete.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The transport-information owner controls published travel guidance, while estates, accessibility and security owners control current site routes, closures and assistance arrangements. Current official disruption notices and competent accessibility decisions take precedence over historical maps, crowd-sourced routes or the May 2021 floor plan. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show any unresolved closure, gradient, lift, lighting or assistance conflict and withhold a definitive route. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Comparison and Diagnosis. Decision interval. Assemble route options at each publication refresh and before a major event, checking current opening, service times, step-free continuity and assistance requirements. Minimum evidence. Require current authoritative notices for every critical segment and at least one accessible alternative where one is officially available. Method and uncertainty. Use rule-based route validation with transfer and disruption checks; show unsupported segments and freshness explicitly. Safe failure. Provide source contacts and state that no verified route can be offered.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose disability, personal destination, journey history or assistance arrangements, promise savings or target delivery, certify accounts, compliance or performance, authorise travel decisions, access permission or emergency routing, or control doors, lifts or transport services. Authority. Only competent owners can confirm accessibility and safety.

WHY THIS IS USEFUL

Keeps visitor guidance current, accessible and honest about closures, outages and route uncertainty.

SU-036CATEGORY 4TRAVEL, ACCESS AND LOWER-IMPACT CHOICES

STAKEHOLDER QUESTION

For a planned session or event, which lower-impact travel options operate at the required times and capacities, including accessible alternatives?

SENSORS AND TELEMETRY

Evidence proposed. Event schedule, aggregate expected attendance bands, public transport timetables, cycle capacity and accessible-route status. Coverage. Take a source snapshot before communication and refresh near the event. Aggregation. Aggregate feasible option ranges by arrival or departure window. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Service disruption, weather, demand uncertainty and first- or last-mile barriers limit feasibility.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Event owners control attendance bands; transport and accessibility owners control service and route status. Current operator and competent accessibility information take precedence over generic journey planners. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; capacity or accessibility conflicts block a definitive option. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Estimate, Forecast, Comparison and Recommendation. Decision interval. Screen time, capacity and accessibility constraints for each option at planning and pre-event communication points. Minimum evidence. Require a confirmed event window, demand range and verified critical-link status. Method and uncertainty. Use multi-constraint option filtering with capacity bands; express demand and disruption uncertainty. Safe failure. Publish a qualified set or no verified option when evidence is incomplete.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose attendee identity, home location or personal mobility needs, promise savings or target delivery, certify accounts, compliance or performance, authorise travel requirements, incentives or event format, or control transport, booking or access systems. Authority. Users retain choice and authoritative travel advice prevails.

WHY THIS IS USEFUL

Allows event organisers to publish feasible lower-impact options early and identify where capacity or accessibility support is needed.

4. Travel, access and lower-impact choices

Category questions 7-8 of 10

SU-037CATEGORY 4TRAVEL, ACCESS AND LOWER-IMPACT CHOICES

STAKEHOLDER QUESTION

Which deliveries or collections could be consolidated, retimed or rerouted to reduce trips and idling without breaching service, safety or supplier constraints?

SENSORS AND TELEMETRY

Evidence proposed. Approved delivery-window bookings, aggregate carrier and vehicle-class counts, loading-bay occupancy and dwell time. **Coverage.** Report daily and weekly without plate, face or driver tracking. **Aggregation.** Aggregate trip, dwell and consolidation opportunity bands by service type. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Unbooked couriers, shared destinations, urgent deliveries and service constraints limit the picture.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement and logistics owners control contracts, delivery rules and service requirements; building operations owns loading-bay status. Signed contractual and operational constraints take precedence over consolidation estimates. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; conflicting carrier, contract or bay records require owner confirmation. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Estimate, Comparison and Recommendation. **Decision interval.** Cluster compatible deliveries and test alternative windows monthly and before contract or campus logistics changes. **Minimum evidence.** Require a representative month, service constraints and verified bay capacity. **Method and uncertainty.** Use time-window clustering with no-disruption constraints; express uncertainty in unbooked trips and supplier participation. **Safe failure.** Propose a limited trial rather than claim full consolidation potential.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose drivers, recipients, suppliers' confidential order details or exact tracking, promise savings or target delivery, certify accounts, compliance or performance, authorise supplier changes, delivery rules or bay allocation, or control gates or logistics platforms. **Authority.** Security, service continuity and urgent needs take precedence.

WHY THIS IS USEFUL

Creates an owner-negotiated delivery trial with measurable trip, idling and service safeguards.

SU-038CATEGORY 4TRAVEL, ACCESS AND LOWER-IMPACT CHOICES

STAKEHOLDER QUESTION

Does pre-arrival guidance let a first-time visitor choose a lower-impact and accessible route without relying on outdated site or building information?

SENSORS AND TELEMETRY

Evidence proposed. Page-view and task-completion aggregates, accessibility status, broken-link checks and voluntary feedback. **Coverage.** Collect without cross-site tracking and summarise weekly. **Aggregation.** Aggregate content reach, successful task signals and unresolved information gaps. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Views do not prove travel choice, feedback is self-selected and accessibility status may change.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Communications owns published guidance; accessibility and transport owners own route and service facts. Current official travel and access information takes precedence over copied or historical content. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; conflicting publication and operational status triggers immediate correction by owners. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Comparison, Diagnosis and Recommendation. **Decision interval.** Test content freshness, findability and task completion monthly and after route or service changes. **Minimum evidence.** Require defined user tasks, anonymous sample thresholds and a manual accessibility check. **Method and uncertainty.** Use funnel and task analysis with qualitative issue coding; express sampling and attribution uncertainty. **Safe failure.** Report engagement signals only, not claimed mode change.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose visitor identity, browsing history, disability or journey choice, promise savings or target delivery, certify accounts, compliance or performance, authorise travel policy or access assurance, or control web, booking or access systems. **Authority.** Do not present a route as usable unless current authoritative status supports it.

WHY THIS IS USEFUL

Improves first-visit route guidance and triggers correction where transport or access information is stale or unverified.

4. Travel, access and lower-impact choices

Category questions 9-10 of 10

SU-039CATEGORY 4TRAVEL, ACCESS AND LOWER-IMPACT CHOICES

STAKEHOLDER QUESTION

For a planned event, what combined travel and building-energy ranges follow from in-person, hybrid and remote formats under stated attendance scenarios?

SENSORS AND TELEMETRY

Evidence proposed. Historical event energy profiles, aggregate attendance bands, room service demand and voluntary anonymous travel-mode assumptions. Coverage. Use event- level totals and comparable seasonal cases. Aggregation. Aggregate scenario ranges for building energy, travel activity and operational carbon. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Remote energy, rebound effects, participation and self-selection are often unmeasured.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Event owners define delivery options; energy, travel and reporting owners control boundaries and factors. Approved scenario methods and current service constraints take precedence over simple averages. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; format, factor or boundary conflicts require transparent alternatives. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Estimate, Forecast and Comparison. Decision interval. Build event-format scenarios at the planning gate using stated attendance ranges, broad origin bands, venue service profiles and remote-platform assumptions. Minimum evidence. Require a clear counterfactual, privacy-safe origin evidence, equivalent event duration and explicit inclusion boundaries for travel and building energy. Method and uncertainty. Use separate travel and building calculations with sensitivity to attendance, mode share, heating or cooling season and remote-energy scope; report ranges rather than a single total. Safe failure. Present the domains separately when boundaries or origin evidence are not comparable.

ANSWER BOUNDARY

Service boundary. Do not collect personal addresses or journeys, identify attendees, infer preferences, claim that one format is universally lower carbon, certify an event footprint, authorise delivery format or travel policy, or control room and digital systems. Authority. Event owners, accessibility needs, educational or operational purpose and the authorised carbon method retain authority.

WHY THIS IS USEFUL

Supports event-format decisions with transparent travel, building-service and uncertainty ranges rather than a single carbon claim.

SU-040CATEGORY 4TRAVEL, ACCESS AND LOWER-IMPACT CHOICES

STAKEHOLDER QUESTION

Did a travel intervention change aggregate mode or trip outcomes beyond seasonality, weather, timetable and sampling effects?

SENSORS AND TELEMETRY

Evidence proposed. Aggregate intervention exposure, repeated anonymous mode counts, facility-use bands and transport disruption context. Coverage. Collect before and after with a comparison period where feasible. Aggregation. Aggregate change ranges by broad cohort or site period. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Seasonality, self-selection, concurrent campaigns and missing baseline data weaken attribution.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The travel-plan owner defines the intervention and outcome; data owners control permitted aggregates. Pre-specified measures and verified counts take precedence over anecdotal success. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; method, exposure or comparison conflicts must be documented. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Observation, Calculation, Estimate, Comparison and Diagnosis. Decision interval. Estimate aggregate change after the intervention and at the next comparable period, separating exposure, mode share and trip volume where evidence allows. Minimum evidence. Require a documented intervention, pre-specified outcomes, adequate baseline and follow-up samples, and records of weather, timetable, access and competing changes. Method and uncertainty. Use weighted before-after analysis, a comparison group or interrupted time series where feasible; show confidence intervals and sensitivity to non-response and seasonal composition. Safe failure. Report association or reach only when causal evidence is insufficient.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose individual travel choices, attitudes or compliance, promise savings or target delivery, certify accounts, compliance or performance, authorise continuation, incentives or policy, or control travel or access systems. Authority. Causal language requires a credible design and transparent assumptions.

WHY THIS IS USEFUL

Shows whether to continue, redesign or stop a travel measure and what stronger evaluation design is needed next time.

5. Water demand, leaks and conservation

Category questions 1-2 of 10

SU-041CATEGORY 5WATER DEMAND, LEAKS AND CONSERVATION

STAKEHOLDER QUESTION

What whole-building water balance is supported by the fiscal supply and verified submetered uses, and what volume remains unallocated or estimated?

SENSORS AND TELEMETRY

Evidence proposed. Main and branch water meters with pulse or digital outputs at the incoming supply, plant amenities and major processes. Coverage. Sample every 1-15 minutes and retain pulse, low-flow and health indicators. Aggregation. Aggregate hour, day and month totals by verified meter hierarchy. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Low-flow resolution, pulse loss, reverse flow and unmapped branches limit allocation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The water and energy manager owns the meter hierarchy; utility bills and fiscal reads support closure. Verified fiscal meters and current plumbing maps take precedence over local estimates. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; meter, billing or mapping conflicts require owner resolution. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Observation, Calculation, Estimate and Comparison. Decision interval. Reconcile main and branch totals and quantify allocated and unaccounted water daily with monthly closure. Minimum evidence. Require valid units, verified meter identities and at least 90% interval completeness. Method and uncertainty. Use water-balance calculation with gap and residual flags; express meter tolerance and an explicit unaccounted range. Safe failure. Return an incomplete balance when branch coverage is insufficient.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose individual water use, account credentials or sensitive process demand, promise savings or target delivery, certify accounts, compliance or performance, authorise billing corrections, metering work or conservation measures, or control meters, pumps or valves. Authority. No flow balance may replace a competent leak or plumbing assessment.

WHY THIS IS USEFUL

Supports a reconciled water position and reveals the submetering or mapping gap that most weakens conservation decisions.

SU-042CATEGORY 5WATER DEMAND, LEAKS AND CONSERVATION

STAKEHOLDER QUESTION

What verified flow persists overnight or during closure, how much is essential, and what residual merits leak investigation?

SENSORS AND TELEMETRY

Evidence proposed. High-resolution main and branch flow meters covering declared low-use periods. Coverage. Sample every 1-5 minutes and retain minimum continuous- flow runs. Aggregation. Aggregate night-flow bands by branch and closure type. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Tank filling, cleaning, cooling, humidification, irrigation and meter resolution can resemble leakage.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The water asset owner controls meter and isolation maps; cleaning, process and plant owners define authorised low-use activity. Verified operational exceptions take precedence over a generic zero-flow expectation. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; conflicting flow, schedule or branch evidence must be checked by those owners. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Observation, Calculation, Comparison, Diagnosis and Recommendation. Decision interval. Detect sustained low-flow events and compare with known essential uses daily with urgent escalation for large changes. Minimum evidence. Require healthy meters, a valid closure window and a current exception list. Method and uncertainty. Use run-length detection and branch balance; express flow range and alternative authorised uses. Safe failure. Issue a possible leak or undocumented-use flag for competent inspection.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose occupant presence, washroom behaviour or security-sensitive schedules, promise savings or target delivery, certify accounts, compliance or performance, authorise isolation, plumbing work or emergency response, or control valves or pumps. Authority. A flow pattern alone does not confirm a leak.

WHY THIS IS USEFUL

Creates a bounded overnight-flow investigation list while protecting essential, hygiene and specialist water services.

5. Water demand, leaks and conservation

Category questions 3-4 of 10

SU-043CATEGORY 5WATER DEMAND, LEAKS AND CONSERVATION

STAKEHOLDER QUESTION

Has water flow increased enough and for long enough to meet the agreed burst or leak escalation rule, and how precisely can the affected area be bounded?

SENSORS AND TELEMETRY

Evidence proposed. High-frequency flow or pressure at the verified main and, where installed, zonal submeters, with valve-state and meter-health information. **Coverage.** Sample at 1-60 seconds where the meter and network support burst detection, otherwise use the fastest validated interval without inventing precision. **Aggregation.** Aggregate threshold duration, rate of rise, sustained excess and the smallest supported isolation area. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Meter latency, tanks, flushing, fire systems, process demand and pressure transients can resemble a leak.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Estates plumbing and incident-response owners control isolation and notification procedures. Verified alarms, inspections and utility notices take precedence over analytics. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; alarm, sensor or incident- log conflicts follow the authoritative response procedure. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Observation, Calculation, Diagnosis and Recommendation. **Decision interval.** Apply only owner-approved burst and leak rules continuously, with immediate routing to the responsible operational channel when thresholds pass. **Minimum evidence.** Require healthy meter status, a validated normal range, aligned clocks and exclusion of known high-flow activities where possible. **Method and uncertainty.** Use rate, persistence and multi-point consistency checks; report confidence and the smallest defensible area rather than a precise source. **Safe failure.** Issue an unverified high-flow alert and direct competent checking when location or cause is uncertain.

ANSWER BOUNDARY

Service boundary. Do not identify people, infer misuse or blame, disclose sensitive isolation details broadly, declare a burst as fact, authorise isolation or entry, or control valves and pumps. **Authority.** Emergency and facilities procedures, competent staff and life-safety water requirements retain precedence.

WHY THIS IS USEFUL

Supports rapid escalation to the responsible water and facilities teams with a clear evidence-based location bound.

SU-044CATEGORY 5WATER DEMAND, LEAKS AND CONSERVATION

STAKEHOLDER QUESTION

Which end uses dominate water demand, and where would permanent or temporary metering reduce decision uncertainty most?

SENSORS AND TELEMETRY

Evidence proposed. Branch meters for washrooms, catering, cooling, irrigation and opt-in process areas, plus fixture counts and plant run-state. **Coverage.** Aggregate daily and monthly with representative spot measurements for unmetered branches. **Aggregation.** Aggregate end-use shares and residual water by service category. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Shared branches, intermittent use, low-flow resolution and changing occupancy limit allocation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The water manager owns meter mapping; service and process owners declare end uses and constraints. Verified physical mapping and direct meters take precedence over model allocations. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; classification, mapping or total-balance conflicts remain visible. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Observation, Calculation, Estimate, Comparison and Recommendation. **Decision interval.** Reconcile measured branches and estimate only the residual categories monthly. **Minimum evidence.** Require a closed main balance and verified coverage of major branches. **Method and uncertainty.** Use hierarchical allocation using measured evidence first; express ranges for every estimated category and the residual. **Safe failure.** Recommend targeted metering when estimation dominates.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose individual hygiene, consumption or service-user identity, promise savings or target delivery, certify accounts, compliance or performance, authorise metering, plumbing or conservation changes, or control fixtures, valves or processes. **Authority.** Estimated end-use shares must remain labelled as estimates.

WHY THIS IS USEFUL

Directs permanent or temporary metering to the end use where it will most improve leak, conservation or recharge decisions.

5. Water demand, leaks and conservation

Category questions 5-6 of 10

SU-045CATEGORY 5WATER DEMAND, LEAKS AND CONSERVATION

STAKEHOLDER QUESTION

Which fixture groups show repeated leak, run-on or high-volume signatures and should be inspected first?

SENSORS AND TELEMETRY

Evidence proposed. Fixture-level run-time or flow logging where technically and legally suitable, supported by periodic non-imaging inspections and maintenance tickets. Coverage. Sample event duration and volume at the fixture or small bank level, avoiding audio, video and user identifiers. Aggregation. Aggregate repeated run-on, refill, drip-equivalent flow and high-volume events by fixture group and week. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Shared pipework, cleaning, sensor fouling, user-activated duration and intermittent faults can create false flags.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Estates plumbing owns fixture inventory and maintenance history; accessibility owners define relevant requirements. Competent inspection and current fixture settings take precedence over automated pattern labels. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; sensor, fixture or maintenance conflicts require site confirmation. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Observation, Comparison, Diagnosis and Recommendation. Decision interval. Detect repeated duration or refill anomalies against fixture-specific baselines daily with a weekly maintenance queue. Minimum evidence. Require a validated normal pattern, sufficient events and no active work. Method and uncertainty. Use event-duration classification with branch-flow corroboration; express confidence and false-positive risk. Safe failure. Create a fixture check task, not a fault certificate.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose washroom use, identity, health or hygiene behaviour, promise savings or target delivery, certify accounts, compliance or performance, authorise fixture changes or remedial work, or control flush valves, taps or isolation valves. Authority. Accessibility, safeguarding and cleaning requirements remain authoritative.

WHY THIS IS USEFUL

Prioritises fixture inspections by persistence and plausible volume while avoiding claims about individual users.

SU-046CATEGORY 5WATER DEMAND, LEAKS AND CONSERVATION

STAKEHOLDER QUESTION

Are cooling, humidification, irrigation or specialist-process water uses proportionate to verified operating state and approved service need?

SENSORS AND TELEMETRY

Evidence proposed. Dedicated flow meters, make-up water, blowdown, conductivity or plant run-state limited to owner-approved assets. Coverage. Sample every 1-15 minutes and align with operating cycles. Aggregation. Aggregate water intensity by process output, runtime or seasonal driver. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Evaporation, treatment cycles, shared meters and unmeasured reject water limit completeness.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The relevant process owner controls intended operation and permissions; water management owns metering. Manufacturer, safety and water-treatment requirements take precedence over generic efficiency targets. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; process, meter or requirement conflicts go to the competent owner. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Calculation, Comparison and Diagnosis. Decision interval. Calculate process-specific water intensity and detect sustained deviation weekly or seasonally as appropriate. Minimum evidence. Require verified returns or output, mapped meters and a documented service baseline. Method and uncertainty. Use normalised trend analysis with control-mode context; express range for output and unmeasured flows. Safe failure. State unknown when service output or make-up balance is missing.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose confidential process details, research content or operator performance, promise savings or target delivery, certify accounts, compliance or performance, authorise process settings, treatment changes or maintenance, or control plant, irrigation or treatment controls. Authority. Water quality, safety and equipment protection take precedence.

WHY THIS IS USEFUL

Supports plant and process checks where water use is inconsistent with verified operating state or service need.

5. Water demand, leaks and conservation

Category questions 7-8 of 10

SU-047CATEGORY 5WATER DEMAND, LEAKS AND CONSERVATION

STAKEHOLDER QUESTION

After adjusting for service intensity, weather and known changes, is water use structurally improving, worsening or inconclusive?

SENSORS AND TELEMETRY

Evidence proposed. Main and branch demand, approved operating schedules, anonymous aggregate use indicators and weather or irrigation context. Coverage. Use daily and monthly data over a representative baseline. Aggregation. Aggregate normalised water predictions and residuals. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Proxy error; visitor variation, leaks and structural service changes can confound normalisation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Water management owns meter fitness; space, operations and project owners supply contextual changes. Approved reporting rules and verified project records take precedence over model assumptions. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; context, boundary or meter conflicts require owner confirmation. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Calculation, Estimate and Comparison. Decision interval. Build normalised demand models and test residual change monthly and after interventions. Minimum evidence. Require at least 12 representative weeks with sufficient coverage and documented structural changes. Method and uncertainty. Use regression or matched-period comparison selected for the evidence; express prediction intervals and sensitivity to proxy choice. Safe failure. Report observed change without attribution when the model is unstable.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose individual attendance, hygiene or behaviour, promise savings or target delivery, certify accounts, compliance or performance, authorise target setting, blame or operational changes, or control water systems. Authority. Correlation with occupancy or weather does not establish cause.

WHY THIS IS USEFUL

Provides a like-for-like trend for conservation planning and states when changing use or weak evidence makes the direction inconclusive.

SU-048CATEGORY 5WATER DEMAND, LEAKS AND CONSERVATION

STAKEHOLDER QUESTION

Which meter identity, unit, billing-window or estimate discrepancies block water reporting or recharge?

SENSORS AND TELEMETRY

Evidence proposed. Utility bills, meter reads and submeter totals linked to a verified meter register. Coverage. Reconcile monthly and after meter exchange, rollover or supplier change. Aggregation. Aggregate variance by meter, unit, period and source. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Estimated reads, multipliers, date-window mismatch and leak adjustments can create differences.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Finance owns invoices; water management owns the meter register and dashboard mappings. Validated fiscal records take precedence for formal totals. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; unresolved identity or period differences require joint owner action. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Calculation, Comparison and Diagnosis. Decision interval. Check serials, multipliers, units, date windows and main-to-branch balance before each reporting close. Minimum evidence. Require matched identities and complete evidence for the billing period. Method and uncertainty. Use line-by-line reconciliation with declared tolerances; express source confidence and residual variance. Safe failure. Withhold formal use when the period or meter cannot be matched.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose account credentials or sensitive billing detail, promise savings or target delivery, certify accounts, compliance or performance, authorise invoice corrections, restatement or meter changes, or control financial systems or meters. Authority. The service must not overwrite authoritative records.

WHY THIS IS USEFUL

Prevents reporting, recharge or intervention evaluation from relying on mismatched water meters, units or billing windows.

5. Water demand, leaks and conservation

Category questions 9-10 of 10

SU-049CATEGORY 5WATER DEMAND, LEAKS AND CONSERVATION

STAKEHOLDER QUESTION

Did a water-saving intervention reduce measured demand under comparable conditions without adverse hygiene, maintenance or service outcomes?

SENSORS AND TELEMETRY

Evidence proposed. Before-and-after fixture or branch flow, approved service output and non-identifying fault or service counts. Coverage. Use a defined baseline and outcome window with seasonal context. Aggregation. Aggregate change ranges in water, service output and adverse indicators. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Season, occupancy, maintenance and concurrent changes may confound results.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The intervention owner defines scope and success; water and service owners control verified outcomes. Competent inspections and service standards take precedence over flow reduction alone. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; conflicting demand or service evidence requires joint interpretation. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Observation, Calculation, Estimate, Comparison and Diagnosis. Decision interval. Compare matched periods or a credible comparison group after a stabilisation period and at the next planning cycle. Minimum evidence. Require documented intervention timing, sufficient baseline and service-quality evidence. Method and uncertainty. Use controlled before-and-after or interrupted time series; express confidence intervals and confounder sensitivity. Safe failure. Report association only if causal evidence is inadequate.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose individual use, hygiene or personal feedback, promise savings or target delivery, certify accounts, compliance or performance, authorise rollout, funding or service-standard changes, or control fixtures or controls. Authority. Do not claim success when water falls but verified service indicators worsen.

WHY THIS IS USEFUL

Shows whether to retain, modify or stop a conservation measure after checking adverse service and maintenance indicators.

SU-050CATEGORY 5WATER DEMAND, LEAKS AND CONSERVATION

STAKEHOLDER QUESTION

During an authorised drought or supply constraint, which non-critical water uses could be reduced first, under what safeguards and approvals?

SENSORS AND TELEMETRY

Evidence proposed. Current water balance, storage or tank status, branch flows, asset criticality and authorised service dependencies. Coverage. Refresh during constraint periods at the fastest reliable source interval. Aggregation. Aggregate reduction scenarios by branch, duration and criticality tier. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Unknown dependencies, rapid conditions and incomplete isolation mapping may make scenarios unsafe.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Incident or estates command owns response; service, accessibility and safety owners define protected uses. Official restriction notices and competent decisions take precedence over analytics. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; uncertain dependencies or isolation conflicts block a recommended action. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Estimate, Forecast, Comparison and Recommendation. Decision interval. Rank owner-approved curtailment options against critical-service constraints during preparedness planning and active constraints. Minimum evidence. Require verified branch mapping, protected-use register and current incident authority. Method and uncertainty. Use constrained scenario analysis with minimum- service floors; express range for demand reduction and duration. Safe failure. Return no safe option and escalate when dependencies are unresolved.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose individual use patterns, health information or sensitive resilience detail, promise savings or target delivery, certify accounts, compliance or performance, authorise restrictions, isolation or emergency response, or control valves, pumps or services. Authority. Emergency command and statutory requirements remain authoritative.

WHY THIS IS USEFUL

Provides a competent-owner reduction sequence with explicit hygiene, safety, accessibility and operational safeguards.

6. Waste, materials, reuse and circularity

Category questions 1-2 of 10

SU-051CATEGORY 6WASTE, MATERIALS, REUSE AND CIRCULARITY

STAKEHOLDER QUESTION

What waste mass by authorised stream is supported for the reporting period, and how complete is the chain of custody from building to destination?

SENSORS AND TELEMETRY

Evidence proposed. Contractor weighbridge or lift records, calibrated platform or bin scales at selected streams, collection timestamps and authorised contamination samples without images of contents. **Coverage.** Capture every collection where possible and run representative measurement campaigns for unweighed streams. **Aggregation.** Aggregate mass by stream, collection point and period with measured, estimated and excluded quantities kept separate. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Shared compactors, missed lifts, tare changes, moisture, rejected loads and broken chain-of-custody can distort totals.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The waste manager owns stream definitions and collection points; contractors own certified weights. Contractor weighbridge or signed lift records take precedence over fill- derived estimates. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; stream, location or weight conflicts require waste-manager resolution. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Calculation, Estimate and Comparison. **Decision interval.** Reconcile collection records, measured weights and contractor destinations monthly and at contract checkpoints. **Minimum evidence.** Require stream definitions, matched collection events, known tare or conversion basis and documented treatment of shared or unweighed loads. **Method and uncertainty.** Use mass-balance and chain-of-custody checks; show measured, estimated, rejected and unallocated quantities with uncertainty. **Safe failure.** Report an incomplete stream total and the broken evidence link rather than infer a destination or recovery outcome.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose personal waste, contents, identities or confidential disposal detail, promise savings or target delivery, certify accounts, compliance or performance, authorise collection changes, contracts or compliance conclusions, or control bins, compactors or access controls. **Authority.** Hazardous and confidential streams follow specialist procedures.

WHY THIS IS USEFUL

Supports reporting and contractor challenge using traceable stream weights and clearly labelled gaps in custody evidence.

SU-052CATEGORY 6WASTE, MATERIALS, REUSE AND CIRCULARITY

STAKEHOLDER QUESTION

Which collection points repeatedly overflow, become contaminated, obstruct routes or become inaccessible, and what service response is justified?

SENSORS AND TELEMETRY

Evidence proposed. Non-imaging ultrasonic fill sensors where the container geometry is suitable, plus structured service inspections, cleaning tickets and contamination checks that do not record waste contents. **Coverage.** Sample fill at 15-60 minutes and record each collection, overflow, obstruction or access event with current point identity. **Aggregation.** Aggregate repeated service failures by collection point, stream, day type and week. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Moved bins, irregular waste, compaction, sensor fouling and subjective inspection categories can misstate condition.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The waste-service owner controls collection-point identities and response rules, while cleaning and accessibility owners confirm service and placement constraints. Signed collection or service records and current site checks take precedence over a sensor estimate; the May 2021 floor plan cannot establish current bin location or route. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show conflicts between fill level, collection ticket and site observation; unresolved safety or access issues must be escalated. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Observation, Comparison, Diagnosis and Recommendation. **Decision interval.** Triage service failures daily and assess persistent patterns weekly. **Minimum evidence.** Require a verified current location, at least three comparable events for a pattern or one immediate safety or access concern, and matched collection context. **Method and uncertainty.** Use event-rate and recurrence analysis with collection timing and local activity context; show sensor and inspection disagreement. **Safe failure.** Issue a location-check request without assigning cause when point identity or evidence is uncertain.

ANSWER BOUNDARY

Service boundary. Do not image or inspect contents for analytics, identify or blame disposers, infer individual behaviour, disclose confidential or hazardous material, authorise relocation or collection changes, or control compactors and access. **Authority.** Waste, cleaning, accessibility and safety owners retain authority over the response.

WHY THIS IS USEFUL

Directs cleaning, waste and facilities teams to the recurring service failure, route obstruction or accessibility issue that needs action.

6. Waste, materials, reuse and circularity

Category questions 3-4 of 10

SU-053CATEGORY 6WASTE, MATERIALS, REUSE AND CIRCULARITY

STAKEHOLDER QUESTION

Can container capacity, location and collection frequency be rebalanced to reduce unnecessary collections and liners while maintaining service?

SENSORS AND TELEMETRY

Evidence proposed. Verified bin capacities, non-imaging fill or weight observations, collection timestamps, liner use and missed-service events for general and recyclable streams. Coverage. Measure across at least six representative weeks, including normal teaching and event periods, before changing a route. Aggregation. Aggregate fill at collection, overflow risk, premature lifts, liner demand and travel by collection point. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Moved containers, seasonal peaks, shared points, compaction and incomplete route records constrain optimisation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The waste-contract owner controls service frequencies and route obligations, while cleaning, accessibility and fire-safety owners control local constraints. Current contracts, verified capacity and competent route decisions take precedence over an optimisation result or historical layout. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show any conflict between capacity, route, access, hygiene or fire constraints and block a recommendation until owners resolve it. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Estimate, Forecast, Comparison and Recommendation. Decision interval. Test alternative capacity and frequency combinations at each service-planning point, using at least six representative weeks and a stated overflow-risk limit. Minimum evidence. Require verified point identity, capacity, collection events and enough observations across expected demand conditions. Method and uncertainty. Use constrained scenario comparison for lifts, liners, route effort and service failures; show seasonal and fill-sensor uncertainty. Safe failure. Retain the current arrangement when evidence cannot demonstrate a lower-burden option within service constraints.

ANSWER BOUNDARY

Service boundary. Do not identify disposers, inspect contents, blame occupants or cleaners, promise savings, authorise contract or route changes, move bins, or control compactors and access. Authority. Hygiene, accessibility, fire safety, staff workload and the authorised service contract retain precedence.

WHY THIS IS USEFUL

Supports a bounded container and collection trial that can reduce excess servicing without creating overflow or contamination.

SU-054CATEGORY 6WASTE, MATERIALS, REUSE AND CIRCULARITY

STAKEHOLDER QUESTION

How much avoidable food waste occurs per catered meal or event, and which prevention trial is proportionate and measurable?

SENSORS AND TELEMETRY

Evidence proposed. Kitchen and dining food-waste scales, aggregate meal counts and production batch totals without customer or staff identity. Coverage. Record by service period and report weekly. Aggregation. Aggregate food waste per meal or batch with preparation and plate-waste bands. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Moisture, mixed packaging, missing meal counts and special events limit comparability.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Catering owns production and meal counts; waste management controls stream definition and contractor weights. Calibrated scales and agreed food-waste definitions take precedence over visual estimates. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; meal, scale or stream conflicts require catering and waste-owner agreement. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Calculation, Estimate, Comparison and Recommendation. Decision interval. Calculate matched waste intensity and test repeatable hotspots weekly and by menu cycle. Minimum evidence. Require verified scales, comparable service periods and sufficient meal-count coverage. Method and uncertainty. Use ratio and matched-period analysis with separate preparation and serving stages; express sampling and menu-mix uncertainty. Safe failure. Report an evidence gap where food and meal data cannot be matched.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose customer choice, staff performance, dietary information or individual consumption, promise savings or target delivery, certify accounts, compliance or performance, authorise menu, procurement or service changes, or control kitchen equipment. Authority. Food safety, nutrition and service requirements remain authoritative.

WHY THIS IS USEFUL

Selects a measurable food-waste prevention trial and defines the meal or event denominator needed to judge it fairly.

6. Waste, materials, reuse and circularity

Category questions 5-6 of 10

SU-055CATEGORY 6WASTE, MATERIALS, REUSE AND CIRCULARITY

STAKEHOLDER QUESTION

Which planned purchases can be met from verified internal reuse stock without compromising safety, function or timing?

SENSORS AND TELEMETRY

Evidence proposed. Owner-approved QR or barcode inventory, condition bands and storage-location status without person tracking. Coverage. Update at movement, inspection and disposal-gate events. Aggregation. Aggregate available assets by type, condition, dimensions and authorised location. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Missing labels, duplicate records, hidden damage and outdated storage status limit confidence.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Asset and procurement owners control ownership, condition and availability; safety and specialist owners control suitability. Current physical inspection and verified ownership take precedence over catalogue status. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; condition, ownership or location conflicts block matching. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Comparison and Recommendation. Decision interval. Match authorised demand specifications to available assets at each purchase or disposal gate. Minimum evidence. Require current condition, dimensions, compatibility and responsible-owner approval. Method and uncertainty. Use constraint matching with confidence bands; express condition and availability uncertainty. Safe failure. Return no verified match rather than suggest an unsafe or unsuitable asset.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose staff movement, personal equipment allocation or confidential asset details, promise savings or target delivery, certify accounts, compliance or performance, authorise reuse, transfer, disposal or purchase, or control stores or inventory systems. Authority. Safety, cybersecurity, warranty and data-handling requirements remain authoritative.

WHY THIS IS USEFUL

Avoids unnecessary purchases by matching verified reusable stock to a live need with condition, safety and timing checks.

SU-056CATEGORY 6WASTE, MATERIALS, REUSE AND CIRCULARITY

STAKEHOLDER QUESTION

Which assets proposed for disposal are viable for refurbishment, redeployment, donation or parts recovery after safety and data checks?

SENSORS AND TELEMETRY

Evidence proposed. Asset age, runtime, condition telemetry, fault counts and authorised inspection results. Coverage. Update monthly and at work-order or disposal events. Aggregation. Aggregate candidate status by asset class, condition and likely service need. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Hidden damage, obsolescence, missing maintenance history and future demand uncertainty limit screening.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Asset owners and CMMS owners control condition and history; safety, cybersecurity and data owners define release constraints. Competent inspection and current support status take precedence over age-based scoring. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; condition, support or data-risk conflicts block a circular option. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Comparison, Diagnosis and Recommendation. Decision interval. Screen whole-life options at disposal gates before replacement or disposal decisions. Minimum evidence. Require current condition, expected need and credible refurbishment or recovery evidence. Method and uncertainty. Use scenario ranges for life extension, cost and carbon; express remaining-life and future-use sensitivity. Safe failure. Route unresolved safety or data risk to the competent owner.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose staff performance, user history or sensitive asset data, promise savings or target delivery, certify accounts, compliance or performance, authorise refurbishment, redeployment, parts recovery or disposal, or control equipment or stores. Authority. Responsible owners decide whether an asset is safe, supported and suitable.

WHY THIS IS USEFUL

Keeps usable assets in service longer and routes only suitable items to refurbishment, redeployment, donation or recovery.

6. Waste, materials, reuse and circularity

Category questions 7-8 of 10

SU-057CATEGORY 6WASTE, MATERIALS, REUSE AND CIRCULARITY

STAKEHOLDER QUESTION

Are project waste plans, transfer records and contractor evidence sufficient to verify quantities, destinations, recovery and on-site reuse without double counting?

SENSORS AND TELEMETRY

Evidence proposed. Project weighbridge tickets, skip collection IDs, material transfer notes and QR-coded reuse lots at project boundaries. Coverage. Update at each movement and monthly gate. Aggregation. Aggregate mass by material, destination and verified reuse route. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Mixed loads, third-party sites, estimated volumes and missing transfer documents limit traceability.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The project manager and principal contractor own site records; waste and legal owners control classification and evidence requirements. Signed transfer documentation and verified receiving evidence take precedence over estimates. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; destination, material or quantity conflicts require project-owner resolution. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Calculation, Comparison and Diagnosis. Decision interval. Reconcile material outflow with project waste plans and destination evidence monthly and at completion. Minimum evidence. Require container identity, movement record and receiving evidence for each material class. Method and uncertainty. Use mass balance with confidence grading; express estimated mass and destination uncertainty. Safe failure. Classify unverified outcomes separately from confirmed reuse or recovery.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose commercially sensitive tender detail or personal contractor data, promise savings or target delivery, certify accounts, compliance or performance, authorise contract action, payment, compliance or project acceptance, or control site or contractor systems. Authority. Formal project governance and competent waste professionals prevail.

WHY THIS IS USEFUL

Allows project and waste owners to challenge missing or inconsistent contractor evidence before recovery claims are reported.

SU-058CATEGORY 6WASTE, MATERIALS, REUSE AND CIRCULARITY

STAKEHOLDER QUESTION

Which waste streams require specialist exclusion, access and reporting rules, and what limited aggregate sustainability evidence may safely be used?

SENSORS AND TELEMETRY

Evidence proposed. Container-presence, sealed status and collection timestamps only where approved, with no content sensing or imaging. Coverage. Report compliance- safe counts or weights at protected aggregate level. Aggregation. Aggregate permitted high-level service indicators, not material composition. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Small volumes, classification sensitivity and chain-of-custody rules deliberately constrain visibility.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Specialist waste, safety, information-governance and security owners define boundaries. Legal duties, authorised procedures and chain-of-custody records take precedence over sustainability analytics. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; any classification or permission conflict defaults to the stricter owner-controlled route. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Comparison and Diagnosis. Decision interval. Apply rule-based exclusion and minimum-safe aggregation whenever data enter the service. Minimum evidence. Require current classification, permitted purpose and approved audience. Method and uncertainty. Use deny-by-default controls with disclosure checks; express uncertainty in classification treated conservatively. Safe failure. Return only a permitted aggregate or no answer.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose contents, identities, diagnoses, research material or security information, promise savings or target delivery, certify accounts, compliance or performance, authorise classification, handling or disposal decisions, or control containers or specialist systems. Authority. Never open, image, infer or combine protected waste content.

WHY THIS IS USEFUL

Protects safety, confidentiality and legal handling while defining the small set of aggregate information available for planning.

6. Waste, materials, reuse and circularity

Category questions 9-10 of 10

SU-059CATEGORY 6WASTE, MATERIALS, REUSE AND CIRCULARITY

STAKEHOLDER QUESTION

Which product or supplier groups offer credible packaging reduction, returnable-container or take-back routes that fit operational constraints?

SENSORS AND TELEMETRY

Evidence proposed. Procurement line data, receiving-area packaging weights from authorised sampling, return-container counts and signed take-back records. Coverage. Aggregate by product group and quarter. Aggregation. Aggregate packaging intensity and verified return activity by supplier or category. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Shared deliveries, unweighed packaging and short samples limit precision.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement owns contract terms; receiving and waste teams supply verified sampling and collection evidence. Signed take-back commitments and measured packaging take precedence over marketing statements. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; supplier, contract or measurement conflicts remain visible. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Comparison and Recommendation. Decision interval. Screen product groups for material volume, repeat frequency and contractual feasibility quarterly and before tender design. Minimum evidence. Require representative sampling, current contract terms and a credible return route. Method and uncertainty. Use range-based opportunity scoring; express sampling and supplier- participation uncertainty. Safe failure. Propose an evidence request or limited trial when proof is weak.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose commercially sensitive prices, staff purchases or supplier-confidential data, promise savings or target delivery, certify accounts, compliance or performance, authorise contract terms, supplier selection or procurement action, or control receiving or procurement systems. Authority. Do not treat a supplier claim as verified delivery.

WHY THIS IS USEFUL

Supports procurement or contract changes that reduce packaging while remaining compatible with storage, hygiene and supplier operations.

SU-060CATEGORY 6WASTE, MATERIALS, REUSE AND CIRCULARITY

STAKEHOLDER QUESTION

Did a waste, reuse or recycling intervention reduce residual waste or increase reuse without worsening contamination, workload, safety or service quality?

SENSORS AND TELEMETRY

Evidence proposed. Before-and-after waste weights, contamination sampling, reuse transactions and authorised service-feedback aggregates. Coverage. Use matched periods and aggregate weekly or monthly. Aggregation. Aggregate change in waste, contamination, reuse and adverse indicators. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Seasonal activity, collection changes, small samples and concurrent initiatives can confound results.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The intervention owner defines success; waste, cleaning, procurement and reuse owners control outcome evidence. Verified contractor records and competent service evidence take precedence over campaign claims. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; conflicting waste and service outcomes require joint interpretation. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Observation, Calculation, Estimate, Comparison and Diagnosis. Decision interval. Compare matched periods or use a credible comparison area after stabilisation and at the next decision point. Minimum evidence. Require a documented intervention, sufficient baseline and pre-defined adverse indicators. Method and uncertainty. Use difference-in-differences or interrupted time series where feasible; express confidence ranges and confounder sensitivity. Safe failure. Report association only when causal evidence is inadequate.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose participant identity, attitudes, disposal behaviour or blame, promise savings or target delivery, certify accounts, compliance or performance, authorise rollout, funding or contract changes, or control waste systems. Authority. Do not call an intervention successful when contamination or service burden materially worsens.

WHY THIS IS USEFUL

Shows whether to scale, redesign or stop the intervention after checking residual waste, reuse, contamination, workload and safety together.

7. Procurement, projects and embodied carbon

Category questions 1-2 of 10

SU-061CATEGORY 7PROCUREMENT, PROJECTS AND EMBODIED CARBON

STAKEHOLDER QUESTION

Which shortlisted materials or products have the lowest credible embodied carbon on an equivalent functional unit, quantity, service life and lifecycle-module basis?

SENSORS AND TELEMETRY

Evidence proposed. Product quantities, service-life assumptions, verified Environmental Product Declarations and supplier technical data. Coverage. Capture source versions at option assessment and procurement gates. Aggregation. Aggregate life-cycle module totals per common functional unit. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Jointed streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Missing declarations, differing boundaries, generic data and uncertain service life limit comparability.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The project and procurement owners define quantities and functional requirements; the carbon specialist controls the method. Verified, current declarations and approved generic datasets take precedence over marketing claims. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; unit, boundary or declaration conflicts must be resolved before comparison. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Calculation, Estimate, Comparison and Recommendation. Decision interval. Normalise options at each design or procurement gate to the same function, quantity, performance, service life, study period and declared lifecycle modules. Minimum evidence. Require functional equivalence, current quantities and product-specific declarations where available, with transparent approved substitutions for missing modules. Method and uncertainty. Use lifecycle-module calculation and a clear data hierarchy; show ranges for quantity, service life, declared-data coverage and generic substitution. Safe failure. State that options are not comparable when function, boundary or evidence cannot be aligned.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose tender-sensitive pricing, supplier-confidential detail or unsupported product claims, promise savings or target delivery, certify accounts, compliance or performance, authorise product selection, specification or procurement, or control design or procurement systems. Authority. Formal design, safety, technical and regulatory requirements prevail.

WHY THIS IS USEFUL

Creates a functionally fair embodied-carbon comparison and identifies evidence gaps that prevent a defensible ranking.

SU-062CATEGORY 7PROCUREMENT, PROJECTS AND EMBODIED CARBON

STAKEHOLDER QUESTION

For a failing or inefficient asset, which option - maintain, recommission, refurbish, retrofit or replace - has the best plausible whole-life carbon and service outcome?

SENSORS AND TELEMETRY

Evidence proposed. Asset energy and runtime, condition evidence, maintenance history and option-specific product quantities. Coverage. Use representative operation and update at the asset decision gate. Aggregation. Aggregate operational and embodied carbon scenarios over a common service period. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Jointed streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Hidden defects, future use, life extension and manufacturer data create wide uncertainty.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The asset owner controls condition and service need; the carbon specialist owns the whole-life method. Competent condition evidence and approved product data take precedence over age or efficiency labels. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; condition, service-life or option-boundary conflicts require owner resolution. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Calculation, Estimate, Forecast, Comparison and Recommendation. Decision interval. Compare maintain, recommission, refurbish, retrofit and replace options at the asset decision point over a common service period and required performance. Minimum evidence. Require verified current condition, remaining-life range, intervention quantities, operational-energy evidence and credible end-of-life assumptions. Method and uncertainty. Use whole-life scenario analysis with embodied, operational, maintenance and downtime components; test break-even timing and sensitivity to service growth and decarbonisation. Safe failure. Present unresolved scenarios rather than selecting an option when condition or functional equivalence is unknown.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose staff performance, sensitive asset history or confidential bids, promise savings or target delivery, certify accounts, compliance or performance, authorise asset option, funding or works, or control equipment or controls. Authority. Safety, service continuity and competent engineering decisions take precedence.

WHY THIS IS USEFUL

Supports a whole-life decision on retaining or changing an asset, including service, maintenance and replacement-timing consequences.

7. Procurement, projects and embodied carbon

Category questions 3-4 of 10

SU-063CATEGORY 7PROCUREMENT, PROJECTS AND EMBODIED CARBON

STAKEHOLDER QUESTION

At the current project gate, is the design within its approved carbon budget, and which quantity, specification or supplier changes explain any variance?

SENSORS AND TELEMETRY

Evidence proposed. Versioned bills of quantities, material or product IDs, design energy estimates and approved supplier substitutions. **Coverage.** Update at each design, procurement and construction gate. **Aggregation.** Aggregate budget variance by life-cycle module, package and change event. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Late quantities, provisional products and inconsistent boundaries can make early totals unstable.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The project sponsor owns the agreed budget; quantity, design and carbon specialists own source inputs. The signed baseline and controlled change decisions take precedence over informal estimates. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; scope, quantity, factor or supplier conflicts remain unresolved until formally owned. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Calculation, Estimate, Comparison, Diagnosis and Recommendation. **Decision interval.** Recalculate the carbon budget and attribute controlled variance at every project gate and monthly during delivery. **Minimum evidence.** Require a versioned baseline, current quantities and approved factor set. **Method and uncertainty.** Use variance decomposition by scope, quantity, product and performance; express estimate class, contingency and interaction ranges. **Safe failure.** State that the budget position is indeterminate when major packages are provisional.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose tender-sensitive rates, supplier-confidential data or personal contractor information, promise savings or target delivery, certify accounts, compliance or performance, authorise budget changes, product approval or project decisions, or control design, procurement or site systems. **Authority.** Formal project governance and competent professionals control action.

WHY THIS IS USEFUL

Gives the project team a traceable carbon-budget variance and the design or procurement decisions responsible for it.

SU-064CATEGORY 7PROCUREMENT, PROJECTS AND EMBODIED CARBON

STAKEHOLDER QUESTION

Which supplier sustainability claims are supported by current, product-specific, third-party evidence, and which remain non-comparable or marketing-only?

SENSORS AND TELEMETRY

Evidence proposed. Supplier declarations, certification records, product documents and traceable batch or chain-of-custody evidence where relevant. **Coverage.** Capture validity dates and source versions at tender and delivery. **Aggregation.** Aggregate claim-evidence status by product and claim type. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Marketing language, expired schemes, mass-balance methods and unsupported offsets can obscure meaning.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement and the project carbon specialist control admissible evidence, while the supplier remains responsible for its claim. Current product-specific third-party declarations, certificates and chain-of-custody evidence take precedence over generic brochures, corporate targets or unverified labels. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show expired, self-declared, offset-dependent, non-equivalent or boundary-mismatched evidence and keep the claim unverified. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Comparison, Diagnosis and Recommendation. **Decision interval.** Screen each claim at shortlisting and before award against the exact product, scope, lifecycle stage, unit and validity period. **Minimum evidence.** Require a traceable source, stated method, third-party status where claimed, and comparability to the procurement requirement. **Method and uncertainty.** Use evidence grading and claim-to-requirement mapping; show missing scope, offset dependence and residual uncertainty. **Safe failure.** Classify the claim as unsupported or non-comparable without alleging misconduct.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose confidential tender information or unsupported legal conclusions, promise savings or target delivery, certify accounts, compliance or performance, authorise supplier acceptance, award or rejection, or control procurement systems. **Authority.** The service does not provide legal assurance or convert marketing statements into verified facts.

WHY THIS IS USEFUL

Separates decision-grade supplier evidence from unverified claims and defines the clarification or documentation needed before scoring.

7. Procurement, projects and embodied carbon

Category questions 5-6 of 10

SU-065CATEGORY 7PROCUREMENT, PROJECTS AND EMBODIED CARBON

STAKEHOLDER QUESTION

How do maintenance, consumables, replacement cycles, adaptability, reuse and end-of-life routes change the whole-life carbon of each option?

SENSORS AND TELEMETRY

Evidence proposed. Material schedules, expected maintenance events, consumable quantities, service-life data and end-of-life route evidence. Coverage. Update at design and delivery gates. Aggregation. Aggregate whole-life carbon by stage, event and option. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Future failure rates, service changes, access constraints and recovery assumptions create uncertainty.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The project, maintenance and procurement owners control service assumptions; the carbon specialist owns calculation rules. Current maintenance strategy and approved product evidence take precedence over generic life assumptions. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; service-life, maintenance or recovery-route conflicts require documented owner decisions. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Calculation, Estimate, Forecast and Comparison. Decision interval. Build event-based life-cycle scenarios over a common study period at option selection and handover. Minimum evidence. Require complete maintenance events, replacement logic and declared end-of-life boundaries. Method and uncertainty. Use scenario accumulation with sensitivity to service life and recovery; express ranges for frequency, material quantity and end-of-life allocation. Safe failure. Show unknown modules separately when evidence is absent.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose confidential cost detail or individual maintainer performance, promise savings or target delivery, certify accounts, compliance or performance, authorise maintenance strategy, project option or contract, or control assets or maintenance systems. Authority. Do not claim future recycling or recovery without a credible route and responsible owner.

WHY THIS IS USEFUL

Prevents a low upfront-carbon option from winning when maintenance, short life, consumables or poor end-of-life outcomes dominate.

SU-066CATEGORY 7PROCUREMENT, PROJECTS AND EMBODIED CARBON

STAKEHOLDER QUESTION

Are procurement sustainability requirements measurable, proportionate, scored consistently and supported by evidence suppliers can realistically provide?

SENSORS AND TELEMETRY

Evidence proposed. Structured requirement fields, supplier evidence templates and product identifiers. Coverage. Check completeness at specification, tender and clarification gates. Aggregation. Aggregate requirement testability, evidence availability and comparison readiness. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. No physical sensor is normally needed before delivery, while heterogeneous supplier documents may not be equivalent.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Procurement owns the specification and evaluation process; legal and sustainability owners define mandatory evidence. The approved procurement method and published tender documents take precedence over informal scoring ideas. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; ambiguous, conflicting or disproportionate requirements require owner correction. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Comparison, Diagnosis and Recommendation. Decision interval. Test each requirement for a measurable unit, boundary, evidence source and decision rule before publication and during clarification. Minimum evidence. Require a defined functional need and an achievable verification route. Method and uncertainty. Use requirement traceability and supplier-burden assessment; express uncertainty about market availability and evidence quality. Safe failure. Convert weak requirements into information requests rather than hidden pass-fail rules.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose undisclosed selection criteria, confidential bids or legal conclusions, promise savings or target delivery, certify accounts, compliance or performance, authorise tender rules, scoring or award, or control procurement platforms. Authority. Equal-treatment, transparency and legal requirements remain authoritative.

WHY THIS IS USEFUL

Improves tender clarity, supplier comparability and scoring assurance without imposing evidence that the market cannot reasonably provide.

7. Procurement, projects and embodied carbon

Category questions 7-8 of 10

SU-067	CATEGORY 7	PROCUREMENT, PROJECTS AND EMBODIED CARBON
STAKEHOLDER QUESTION		
Which refrigerant-containing systems create the highest potential climate impact when verified charge, GWP, leak history, efficiency and replacement timing are combined?		
SENSORS AND TELEMETRY	Evidence proposed. Approved refrigerant inventory, leak-sensor flags where fitted, pressure or temperature diagnostics, runtime and service records. Coverage. Capture alarms at source intervals and summarise annually or after service. Aggregation. Aggregate charge, verified loss events and scenario climate impact by asset. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Sensor absence, false alarms, unknown charge and unverified service reports limit precision.	
AUTHORITATIVE SUPPORTING DATA	Owners and precedence. The refrigeration asset owner and competent contractor control charge and leak records; the reporting owner controls factors. Signed service documentation and current refrigerant factors take precedence over inferred loss. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; charge, asset or service-history conflicts require competent confirmation. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.	
ANALYSIS REQUIRED	Operation type. Lookup, Observation, Calculation, Estimate, Comparison and Recommendation. Decision interval. Screen direct climate-impact exposure annually and at major service or replacement points, combining verified charge, current GWP, signed leakage history and feasible replacement pathways. Minimum evidence. Require verified asset identity, refrigerant type and charge or a bounded owner-approved estimate, plus current factor and service records. Method and uncertainty. Use scenario ranges for leakage, recharge, recovery, efficiency and replacement timing; show wide uncertainty where charge or loss is not measured. Safe failure. Route suspected leakage through competent safety procedures and do not quantify it as fact.	
ANSWER BOUNDARY	Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose restricted plant detail or contractor performance, promise savings or target delivery, certify accounts, compliance or performance, authorise refrigerant work, replacement or compliance action, or control refrigeration or heat-pump systems. Authority. Competent refrigerant specialists and statutory duties take precedence.	
WHY THIS IS USEFUL	Prioritises competent inspection and replacement planning for assets with the largest plausible direct and operational climate exposure.	

SU-068	CATEGORY 7	PROCUREMENT, PROJECTS AND EMBODIED CARBON
STAKEHOLDER QUESTION		
How should ICT equipment options be compared across operational energy, embodied carbon, performance, security, service life, maintainability and reuse?		
SENSORS AND TELEMETRY	Evidence proposed. Approved product test data, representative plug-load measurements, asset inventory and authorised refurbishment or reuse history. Coverage. Aggregate by device class and update at procurement cycles. Aggregation. Aggregate whole-life scenario ranges per service function. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. User workload, vendor modes, rapidly changing models and missing declarations limit comparison.	
AUTHORITATIVE SUPPORTING DATA	Owners and precedence. IT and equipment owners define functional and security needs; procurement and carbon owners control evidence and method. Verified test standards, warranties and asset- security requirements take precedence over simple label comparisons. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; functional, security or product-evidence conflicts block ranking. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.	
ANALYSIS REQUIRED	Operation type. Lookup, Observation, Calculation, Estimate, Forecast, Comparison and Recommendation. Decision interval. Compare functionally and security-equivalent equipment options before purchase, redeployment and refresh over a common service period. Minimum evidence. Require representative energy evidence, credible product or generic embodied data, performance requirements, service-life assumptions and reuse or disposal constraints. Method and uncertainty. Use whole-life scenarios with workload, power-management, maintenance, reuse and refresh alternatives; show uncertainty from workload and rapidly changing models. Safe failure. State that options are non-equivalent when required performance, security or evidence cannot be matched.	
ANSWER BOUNDARY	Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose device content, network use, user activity or confidential security controls, promise savings or target delivery, certify accounts, compliance or performance, authorise purchase, redeployment or disposal, or control IT equipment or networks. Authority. Cybersecurity, data protection and service requirements remain authoritative.	
WHY THIS IS USEFUL	Supports equipment standards, life extension and reuse decisions that meet performance, security and service requirements.	

7. Procurement, projects and embodied carbon

Category questions 9-10 of 10

SU-069CATEGORY 7PROCUREMENT, PROJECTS AND EMBODIED CARBON

STAKEHOLDER QUESTION

Has project handover supplied verified meter and point mapping, commissioning, controls narratives, asset data and baselines needed for performance verification?

SENSORS AND TELEMETRY

Evidence proposed. Commissioning test results, meter-health tests, point lists, asset IDs and as-built digital handover evidence. **Coverage.** Capture at completion and update through the defects period. **Aggregation.** Aggregate handover completeness and test status by system and sustainability outcome. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Temporary sensors, undocumented substitutions and incomplete as-built records reduce usability.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The project manager and asset-information owner control handover requirements; commissioning specialists own signed tests. Signed commissioning and verified as- built evidence take precedence over design intent. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; asset, meter or document conflicts remain open until the responsible owner resolves them. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Comparison, Diagnosis and Recommendation. **Decision interval.** Trace each sustainability outcome to an asset, meter, source and acceptance test at handover and closeout. **Minimum evidence.** Require unique asset identities, verified mappings and complete mandatory evidence. **Method and uncertainty.** Use requirements traceability and sample field checks; express coverage confidence and unresolved-item severity. **Safe failure.** Do not close the evidence route where critical mappings or tests are missing.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose commercially sensitive contract detail or personal contractor data, promise savings or target delivery, certify accounts, compliance or performance, authorise project acceptance, payment or formal completion, or control assets, meters or project systems. **Authority.** Formal acceptance remains with authorised project governance.

WHY THIS IS USEFUL

Creates a handover defect list before the project team disperses and protects the evidence needed for later performance verification.

SU-070CATEGORY 7PROCUREMENT, PROJECTS AND EMBODIED CARBON

STAKEHOLDER QUESTION

What energy, operational-carbon, water, waste and embodied-carbon outcomes are plausible for each project option, and which material trade-offs remain unresolved?

SENSORS AND TELEMETRY

Evidence proposed. Versioned design models, material quantities, proposed equipment performance, water estimates and waste plans. **Coverage.** Update at option and design gates with source-specific validity. **Aggregation.** Aggregate multi-domain scenario ranges with interactions and trade-offs. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Future use, controls, climate, construction uncertainty and incompatible boundaries can dominate results.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The project sponsor owns the decision scope; domain specialists control methods and authoritative inputs. Approved project requirements and domain-specific methods take precedence over a single composite score. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; boundary, unit or trade-off conflicts must be displayed for decision-makers. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Calculation, Estimate, Forecast, Comparison and Recommendation. **Decision interval.** Run coordinated domain scenarios with common service assumptions at major option gates. **Minimum evidence.** Require a defined counterfactual, equivalent service and documented domain boundaries. **Method and uncertainty.** Use separate-domain calculations plus interaction checks; express ranges and sensitivity for each outcome rather than forced aggregation. **Safe failure.** Present unresolved trade-offs instead of declaring one overall winner.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose tender-sensitive detail, unsupported health claims or user behaviour, promise savings or target delivery, certify accounts, compliance or performance, authorise project approval, option selection or budget, or control design or construction systems. **Authority.** Competent professional judgement and formal governance control the decision.

WHY THIS IS USEFUL

Lets project sponsors compare multi-domain outcomes and escalate unresolved trade-offs before option selection or approval.

8. Targets, baselines, reporting, scenarios and engagement

Category questions 1-2 of 10

SU-071CATEGORY 8TARGETS, BASELINES, REPORTING, SCENARIOS AND ENGAGEMENT

STAKEHOLDER QUESTION

Is the formal target baseline complete, representative, reproducible and approved, or does it require correction, restatement or replacement?

SENSORS AND TELEMETRY

Evidence proposed. Reconciled utility and submeter data, meter-health records and known operational-change flags. Coverage. Use the approved baseline period and retain source versions. Aggregation. Aggregate coverage, completeness and representativeness indicators by target metric. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Closures, projects, meter exchange and missing periods can make a year unrepresentative.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The reporting owner approves the baseline; energy, water and finance owners supply reconciled evidence. Approved methodology and finance-aligned actuals take precedence over dashboard convenience. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; period, boundary or source conflicts must be resolved before target use. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Calculation, Comparison, Diagnosis and Recommendation. Decision interval. Test completeness, representativeness, structural stability and reproducibility before baseline approval and at each material boundary or meter change. Minimum evidence. Require documented boundaries, owner-approved measured or estimated coverage, explained closures and projects, and a reproducible source-to-output path; no universal percentage overrides the approved method. Method and uncertainty. Use coverage, change-point, holdout and sensitivity tests; show how baseline choice and approved estimation affect the target path. Safe failure. Declare the baseline unsuitable and identify the smallest correction or alternative period needed.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose individual activity, confidential accounts or unsupported causal explanations, promise savings or target delivery, certify accounts, compliance or performance, authorise baseline approval, target setting or restatement, or control source records. Authority. Formal reporting methods and authorised owners remain decisive.

WHY THIS IS USEFUL

Protects target tracking from an unstable baseline and records the governance decision needed for correction or restatement.

SU-072CATEGORY 8TARGETS, BASELINES, REPORTING, SCENARIOS AND ENGAGEMENT

STAKEHOLDER QUESTION

Against the approved target, what change remains after separating boundary, factor, weather, service and method effects from actual performance?

SENSORS AND TELEMETRY

Evidence proposed. Versioned KPI inputs, reconciled utilities, waste records and approved factors. Coverage. Update monthly or quarterly with explicit coverage. Aggregation. Aggregate actual, method-change and boundary-change components of target variance. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Late invoices, factor updates, reporting changes and missing evidence can shift apparent progress.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The reporting owner controls targets, boundaries and methods; finance, energy, water and waste owners supply authoritative actuals. Approved restatements and current methods take precedence over prior dashboard values. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; unresolved method, boundary or source differences must be disclosed. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Calculation, Estimate, Comparison and Diagnosis. Decision interval. Calculate target variance at each approved interval and bridge reported change into actual use, weather or service context, boundary, factor, estimation and method components. Minimum evidence. Require reconciled actuals, versioned methods, approved target and boundary definitions, and documented material changes. Method and uncertainty. Use like-for-like restatement only where authorised, with bridge analysis and residual variance; show confidence and any non-comparable period. Safe failure. State that progress is indeterminate rather than mixing accounting effects with operational performance.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose individual contribution, team blame or confidential operational data, promise savings or target delivery, certify accounts, compliance or performance, authorise target claims, restatement or corrective programme, or control reporting or source systems. Authority. Do not mix method effects with operational change.

WHY THIS IS USEFUL

Explains measured progress without confusing method, boundary or weather changes with genuine performance change.

8. Targets, baselines, reporting, scenarios and engagement

Category questions 3-4 of 10

SU-073CATEGORY 8TARGETS, BASELINES, REPORTING, SCENARIOS AND ENGAGEMENT

STAKEHOLDER QUESTION

Which fuels, meters, activities, spaces and supply-chain elements remain excluded, estimated or weakly evidenced within the current boundary?

SENSORS AND TELEMETRY

Evidence proposed. Meter and source registry, asset and space maps, procurement categories and explicit estimation flags. Coverage. Update quarterly and after scope or system changes. Aggregation. Aggregate coverage matrix by emission or resource source and evidence grade. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Unknown assets, shared services and third-party data may leave blind spots.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The reporting owner defines the boundary; source owners confirm current coverage and access. Approved boundary documents take precedence over discovered data availability. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; scope, ownership or source conflicts require governance resolution. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Calculation, Estimate, Comparison and Diagnosis. Decision interval. Map every required source to boundary status, owner, evidence grade, estimation method and permission quarterly and before formal reporting. Minimum evidence. Require a current approved boundary and a source inventory that includes explicit unknown, excluded and third-party dependencies. Method and uncertainty. Use coverage mapping, materiality screening and evidence-strength grading; show estimated and unallocated ranges without treating unavailable data as zero. Safe failure. Publish the gap register and withhold a completeness claim.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose restricted supplier, tenant or research information, promise savings or target delivery, certify accounts, compliance or performance, authorise boundary expansion, data access or reporting inclusion, or control source systems. Authority. Data availability alone must not redefine the approved boundary.

WHY THIS IS USEFUL

Creates a transparent completeness register and prioritises the exclusions or estimates that most affect the reported result.

SU-074CATEGORY 8TARGETS, BASELINES, REPORTING, SCENARIOS AND ENGAGEMENT

STAKEHOLDER QUESTION

Which factor, conversion, allocation and calculation versions produced each reported figure, and what changed after method updates?

SENSORS AND TELEMETRY

Evidence proposed. A versioned factor registry and reproducible calculation logs linked to each output. Coverage. Capture source, scope, unit, validity period and retrieval time for every calculation. Aggregation. Aggregate factor and method lineage by metric and reporting period. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Retroactive source updates and method mismatch can affect comparability.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The reporting owner controls approved methods and factors; authoritative publishers own source values. The factor valid for the stated period and approved method takes precedence over the latest value by default. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; factor, period or method conflicts require controlled resolution. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Calculation, Comparison and Diagnosis. Decision interval. Reproduce each affected output and produce a transparent old-to-new bridge whenever a factor, allocation or method version changes. Minimum evidence. Require versioned inputs, a reproducible formula or code path, stated validity periods and an authorised method decision. Method and uncertainty. Use parallel calculation with variance attribution and materiality checks; preserve both results and explain factor, boundary and allocation effects. Safe failure. Retain both outputs as non-comparable when reproduction or approval is missing.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose confidential calculations beyond the permitted audience, promise savings or target delivery, certify accounts, compliance or performance, authorise method adoption, restatement or formal publication, or control reporting systems. Authority. Do not overwrite approved historical records without authorised governance.

WHY THIS IS USEFUL

Makes every reported figure reproducible and supports controlled recalculation when factors or methods change.

8. Targets, baselines, reporting, scenarios and engagement

Category questions 5-6 of 10

SU-075	CATEGORY 8	TARGETS, BASELINES, REPORTING, SCENARIOS AND ENGAGEMENT
STAKEHOLDER QUESTION		
Which portfolio of approved operational, maintenance, retrofit, renewable and demand actions can plausibly close the target gap within known constraints?		
SENSORS AND TELEMETRY	Evidence proposed. Validated baselines, candidate-project estimates, asset criticality, future-service plans and approved renewable evidence. Coverage. Update at portfolio planning intervals. Aggregation. Aggregate target-gap scenarios by action bundle and delivery period. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Measure interactions, rebound, sequencing, estimate class and future service growth limit certainty.	
AUTHORITATIVE SUPPORTING DATA	Owners and precedence. The target owner controls ambition and boundary; finance, estates, procurement and service owners control costs and constraints. Approved plans, safety requirements and current evidence take precedence over modelled bundles. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; delivery, interaction or estimate conflicts remain visible. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.	
ANALYSIS REQUIRED	Operation type. Lookup, Calculation, Estimate, Forecast, Comparison and Recommendation. Decision interval. Run pathway and portfolio scenarios against target and service constraints annually and after major plan changes. Minimum evidence. Require a reconciled baseline, action definitions, delivery windows and owner-approved constraints. Method and uncertainty. Use scenario optimisation with interaction and double-counting checks; express delivery, saving and cost ranges plus rank stability. Safe failure. Show a residual target gap when no credible bundle closes it.	
ANSWER BOUNDARY	Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose individual contribution, staff performance or concealed service trade-offs, promise savings or target delivery, certify accounts, compliance or performance, authorise projects, budgets, staffing or target commitments, or control plant or portfolio systems. Authority. Do not count mutually exclusive measures twice or present a modelled pathway as guaranteed delivery.	
WHY THIS IS USEFUL	Builds a feasible delivery pathway, exposes dependencies and shows the residual gap rather than assuming all nominal savings add together.	

SU-076	CATEGORY 8	TARGETS, BASELINES, REPORTING, SCENARIOS AND ENGAGEMENT
STAKEHOLDER QUESTION		
Under current trend, planned changes and plausible weather and service scenarios, what range of energy and operational-carbon outcomes should be planned for?		
SENSORS AND TELEMETRY	Evidence proposed. Reconciled recent utilities, approved project schedules, service plans and weather or climate scenarios. Coverage. Refresh monthly and at material plan changes. Aggregation. Aggregate forecast distributions by month, fuel and reporting boundary. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Future weather, service demand, invoice lag and project timing create uncertainty.	
AUTHORITATIVE SUPPORTING DATA	Owners and precedence. Energy and reporting owners control actuals and boundaries; planning and project owners supply current assumptions. Approved plans and reconciled actuals take precedence over informal expectations. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; forecast-input or schedule conflicts require owner confirmation. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.	
ANALYSIS REQUIRED	Operation type. Lookup, Observation, Estimate, Forecast and Comparison. Decision interval. Produce scenario forecasts with trend, weather and planned-change components monthly with annual horizon refresh. Minimum evidence. Require a stable recent baseline, current plans and at least one approved weather scenario. Method and uncertainty. Use probabilistic forecasting and back-testing; express prediction intervals and forecast-bias history. Safe failure. Issue a range and risk flag rather than a single precise forecast.	
ANSWER BOUNDARY	Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose individual activity or undisclosed operational plans, promise savings or target delivery, certify accounts, compliance or performance, authorise spending, target claims or operational changes, or control building services. Authority. Forecasts are not promises and must show assumptions and uncertainty.	
WHY THIS IS USEFUL	Provides a planning range for budgets, operations and target risk under explicit weather, service and delivery assumptions.	

8. Targets, baselines, reporting, scenarios and engagement

Category questions 7-8 of 10

SU-077

CATEGORY 8

TARGETS, BASELINES, REPORTING, SCENARIOS AND ENGAGEMENT

STAKEHOLDER QUESTION

Which carbon-reduction proposals should progress first after combining abatement, cost, risk, service, evidence quality, dependencies and delivery capacity?

SENSORS AND TELEMETRY

Evidence proposed. Verified project estimates, baseline evidence, cost ranges, asset criticality, delivery dependencies and owner readiness. **Coverage.** Refresh at each portfolio cycle. **Aggregation.** Aggregate multi-criteria project position and rank stability. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Early estimates, interaction, optimism bias and delivery constraints can change ordering.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The portfolio owner defines criteria and weights; finance, service and carbon owners control authoritative inputs. Approved business-case and safety evidence take precedence over preliminary scores. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; weight, estimate or dependency conflicts must be presented to decision-makers. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Calculation, Estimate, Comparison and Recommendation. **Decision interval.** Score projects transparently and test sensitivity to weights and uncertain estimates at each portfolio decision. **Minimum evidence.** Require defined criteria, comparable boundaries and a minimum evidence grade. **Method and uncertainty.** Use multi-criteria analysis with dominance and dependency checks; express rank ranges and fragile assumptions. **Safe failure.** Group unstable projects rather than force a precise order.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose commercially sensitive bids, personal performance or hidden value judgements, promise savings or target delivery, certify accounts, compliance or performance, authorise funding, programme order or project approval, or control portfolio or finance systems. **Authority.** The score supports governance; it does not replace it.

WHY THIS IS USEFUL

Creates a portfolio shortlist that balances carbon value with cost, risk, evidence and delivery capacity instead of ranking by headline saving alone.

SU-078

CATEGORY 8

TARGETS, BASELINES, REPORTING, SCENARIOS AND ENGAGEMENT

STAKEHOLDER QUESTION

Did an engagement campaign credibly change an aggregate outcome, and should it continue, be redesigned or stop without surveillance or blame?

SENSORS AND TELEMETRY

Evidence proposed. Aggregate outcome meters, anonymous campaign reach, voluntary feedback and coarse participation counts. **Coverage.** Use pre-defined before and after windows at site or protected-group level. **Aggregation.** Aggregate outcome and reach ranges with contextual changes. **Quality gate.** Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Self-selection, concurrent initiatives, weather and service change can confound attribution.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The campaign owner controls intervention design; domain and privacy owners approve measures and audiences. Verified aggregate outcomes and pre-specified indicators take precedence over anecdotal success. **Time, access and provenance.** Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. **Staleness and conflicts.** Show stale records and uncertain mappings; exposure, outcome or permission conflicts require a restricted or no-answer result. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Observation, Calculation, Estimate, Comparison, Diagnosis and Recommendation. **Decision interval.** Estimate aggregate outcome change after the campaign and at the next planning point, using a credible comparison and pre-defined adverse indicators. **Minimum evidence.** Require a baseline, sufficient privacy-safe sample, documented campaign exposure and context for weather, service and concurrent initiatives. **Method and uncertainty.** Use controlled before-after, interrupted time series or matched comparison where feasible; show confidence intervals and alternative explanations. **Safe failure.** Report reach or association only when attribution is weak.

ANSWER BOUNDARY

Service boundary. Do not identify participants or non-participants, infer personal attitudes or behaviour, compare small groups, blame occupants, promise savings, certify campaign impact, authorise incentives or policy, or control communications and building systems. **Authority.** Privacy owners, domain owners and the campaign decision-maker retain authority.

WHY THIS IS USEFUL

Supports a continue, redesign or stop decision and improves future campaign evaluation without monitoring individuals.

8. Targets, baselines, reporting, scenarios and engagement

Category questions 9-10 of 10

SU-079CATEGORY 8TARGETS, BASELINES, REPORTING, SCENARIOS AND ENGAGEMENT

STAKEHOLDER QUESTION

Which dashboard measures are decision-ready, caveated, directional, restricted or unsuitable, and what owner and caveat should accompany each?

SENSORS AND TELEMETRY

Evidence proposed. Data-quality, freshness, lineage and permission metadata attached to every metric. Coverage. Update with each source refresh and material method change. Aggregation. Aggregate metric status, audience and caveat by dashboard element. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. A visual dashboard can conceal stale, estimated or conflicting evidence.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Metric owners define authoritative sources and audiences; finance, reporting, security and privacy owners control restricted detail. Approved figures and current permissions take precedence over cached dashboard values. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; conflicts between display and source must make the metric unavailable. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Comparison, Diagnosis and Recommendation. Decision interval. Apply evidence-readiness, freshness, permission and audience rules before every metric render and whenever a source or method changes. Minimum evidence. Require current source health, owner-confirmed method version, sufficient coverage, visible uncertainty and a valid audience permission. Method and uncertainty. Use rule-based release with four outcomes - decision-ready, caveated, directional or unavailable - and link each non-ready metric to its owner and reason. Safe failure. Suppress detail or mark the metric unavailable rather than displaying a polished but unsupported number.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose restricted source detail, small-group data or personal information, promise savings or target delivery, certify accounts, compliance or performance, authorise management action, formal reporting or data release, or control source or control systems. Authority. Never hide material uncertainty behind visual polish.

WHY THIS IS USEFUL

Prevents provisional or restricted metrics from being mistaken for approved performance and gives users a clear owner and caveat.

SU-080CATEGORY 8TARGETS, BASELINES, REPORTING, SCENARIOS AND ENGAGEMENT

STAKEHOLDER QUESTION

Which high-value sustainability decisions remain unanswerable, and what smallest verified evidence, permission, mapping or ownership upgrade would unlock them?

SENSORS AND TELEMETRY

Evidence proposed. An evidence-gap registry covering mappings, sensor health, data owners, permissions, temporal coverage and quality limits for all 80 questions. Coverage. Refresh quarterly and after material system changes. Aggregation. Aggregate answerability, value and smallest feasible upgrade by question. Quality gate. Gate use on verified collection method or deployment, physical location and current asset/zone mapping where relevant, units, commissioning, and relevant calibration or functional checks. Joined streams require clock alignment; report uptime or response coverage, latency, gaps, substitutions, drift and raw-to-aggregate lineage. Registry accuracy depends on active owner confirmation and may lag operational change.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Each authoritative owner confirms source and constraint status; the sustainability governance lead prioritises gaps. Current professional, safety and reporting requirements take precedence over automated completeness scores. Time, access and provenance. Record effective, observed and retrieved times, applicable validity or version, role-based permission and source-to-output provenance. Staleness and conflicts. Show stale records and uncertain mappings; owner disagreement remains an unresolved dependency. Name the affected period and resolver; unresolved conflicts remain visible and block decision-ready release.

ANALYSIS REQUIRED

Operation type. Lookup, Observation, Comparison, Diagnosis and Recommendation. Decision interval. Score answerability, decision value, risk and effort quarterly and at roadmap decisions. Minimum evidence. Require an owner-confirmed gap, defined decision and lawful purpose. Method and uncertainty. Use multi-criteria prioritisation with dependency and risk checks; express priority ranges and sensitivity to owner input. Safe failure. Return unknown and identify the smallest safe evidence upgrade.

ANSWER BOUNDARY

Service boundary. Do not identify or track people, infer behaviour or assign blame; do not disclose fabricated answers, missing facts, restricted systems or personal information, promise savings or target delivery, certify accounts, compliance or performance, authorise procurement, access or system deployment, or control sensors, integrations or controls. Authority. Unverified sources and integrations remain proposed.

WHY THIS IS USEFUL

Turns unknowns into a prioritised evidence and governance roadmap, focusing investment on the smallest upgrade that unlocks a real decision.

Summary decision map

The catalogue supports a controlled path from proposed evidence to a bounded team decision.

DOMAIN	TYPICAL DECISION	PRIMARY EVIDENCE	AUTHORITATIVE OWNERS
1. Electricity, gas, heat and carbon	Reconcile totals, detect material patterns and select justified investigations.	Fiscal meters, submeters, bills, factors and service context.	Energy, finance, reporting
2. HVAC and environmental efficiency	Tune schedules, inspect controls and prioritise recommissioning.	BMS states, energy, temperatures, flow, air and approved modes.	Mechanical, controls, service
3. Space use and service intensity	Normalise fairly, test consolidation and plan service-zone changes.	Mapped zone energy, service hours and privacy-safe aggregate use context.	Space, energy, accessibility
4. Travel and lower-impact choices	Plan facilities, guidance, events, deliveries and aggregate interventions.	Anonymous surveys, aggregate counts, charger/fleet totals and current notices.	Transport, estates, events
5. Water	Detect unusual flow, prioritise checks and evaluate conservation.	Fiscal and zonal flow, fixture logging, bills and service records.	Water, facilities, finance
6. Waste, reuse and circularity	Measure streams, fix service failures and choose reuse or take-back actions.	Weights, fill events, collection records, reuse inventory and contracts.	Waste, cleaning, procurement
7. Procurement and projects	Compare options, manage carbon budgets and secure usable handover evidence.	Quantities, declarations, specifications, commissioning and asset data.	Project, procurement, carbon
8. Targets, scenarios and engagement	Track like-for-like progress, plan pathways and release trustworthy metrics.	Reconciled KPIs, versioned methods, plans, scenarios and permissions.	Reporting, finance, governance

VERIFY

Confirm modality, location, mapping, commissioning, checks, clocks and health.

TRACE

Name owner, precedence, times, permission, provenance and unresolved conflicts.

RECONCILE

Resolve identity, unit, period, boundary and allocation before using totals.

COMPARE

Use equivalent function, service, weather, evidence and lifecycle boundaries.

SCENARIO

Show assumptions, ranges, constraints, sensitivities and unresolved trade-offs.

REFUSE SAFELY

Return caveated, directional or unknown when a required gate fails.

Evidence and permissions rules

These gates apply before any catalogue question is answered or released.

1	LAWFUL PURPOSE AND OWNER Name the decision, responsible owner, permitted audience and action boundary.	2	CURRENT SPATIAL TRUTH Verify current asset, service-zone and room mapping. The May 2021 floor plan is historical only.	3	DEPLOYMENT TRUTH Treat every sensor and integration as proposed until method/deployment, mapping, commissioning, permission and health pass.
4	SOURCE PRECEDENCE Use the current authoritative owner and approved method; supporting telemetry cannot silently replace them.	5	FOUR TIMES Record effective, observed, retrieved and decision times, plus factor, method and forecast validity periods.	6	MODALITY AND COVERAGE Confirm physical modality/location, spatial and temporal coverage, sampling and aggregation suit the claim.
7	TECHNICAL QUALITY Check units, direction, calibration or functional verification, clock alignment, uptime, latency, gaps, substitutions and drift.	8	PROVENANCE AND CONFLICTS Preserve raw-to-aggregate lineage, source/version and resolver. Keep unresolved conflicts visible.	9	PRIVACY BY AGGREGATION Use the coarsest lawful evidence; suppress small groups and retain no face, device, credential or journey identity.
10	NO BEHAVIOURAL OR CAUSAL OVERREACH Do not infer behaviour, intent, responsibility or blame, and do not turn weak association into causation.	11	ACCOUNTING, AUTHORITY AND CONTROL Keep approved reporting separate from operational signals. Do not certify, authorise work or budgets, or operate plant.	12	SAFE FAILURE State what is unknown, why it matters, who owns the gap and the smallest safe evidence upgrade.

RELEASE OUTCOMES

DECISION-READY All required evidence, method, permission, quality and authority gates pass.	CAVEATED A bounded limitation is material, explicit and acceptable for the stated action.	DIRECTIONAL Useful only to prompt checking, measurement, option development or owner dialogue.	UNKNOWN / ESCALATE A required source, owner, permission, quality, method or safety condition fails.
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